

SUBSYSTEM ENGINEERING MANAGEMENT
PLAN (SEMP)

Solution Element Virtual Interlocking OBB Phase 1 (VIL OBB 1.0.0)
Contract: Digitales Stellwerk DSTW – Rahmenvereinbarung
Country: Austria

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1. SCOPE

The System Engineering Management Plan (SEMP) describes the organisation, work packages, milestones and application of the Management System DDQS process release number 2025-08.

This SEMP deals with the scope of the price position Basic Virtualization (B-Vi) Phase 1 of the Solution Element Virtual Interlocking OBB Phase 1 (VIL OBB 1.0.0) and contract mentioned in Table 1: Solution Element Identification.

“Basic virtualization” (B-Vi) is the minimum technical requirement to be implemented upon the first delivery of a DSTW including the implementation of the Virtual Interlocking (VIL) and is a subsystem of the Digital Interlocking’s Central Processing system (DIL-CP). A brief description of the subsystem overview can be found in section [1.2].

1.1 IDENTIFICATION

1.1.1 GENERAL

Table 1: Solution Element Identification

Project Name:	DSTW ÖBB Entwicklung
Project ID:	100025795 / ATRELDE91 (CFRD) 100020889 / ATRENDDB35 ¹
Solution Element/ Subsystem:	Solution Element Virtual Interlocking OBB Phase 1 (VIL OBB 1.0.0) ²
Contract:	Digitales Stellwerk DSTW – Rahmenvereinbarung ³
Identification⁴:	LockTrac 6131 ELEKTRA ELEKTRA BL OBB 40.00 ELEKTRA PV GEN 2.20.0 CAR-RED_1.0.0 EEC SW OBB 04.00 OREST 19.50

¹ Please note that some work packages are managed under a separate Project ID described in chapter [1.1.4]

² Please note that at this stage the focus of the development of the Solution Element is on Phase 1 and this is indicated by 1.0.0 – for the phases see [SEMP DIL]

³ Please note that this SEMP Edition is based on the 4th Edition of the offer placed to ÖBB at 16th of June 2025 and the negotiations process is ongoing. The KOM will take place after the contract has been signed. Current activities are based on the customer requirements [CRS]. SOR2 will be repeated after official KOM.

⁴ The identification refers to the key products/components included in the Solution Element as well the product LockTrac 6131 ELEKTRA itself.

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	TAFE 1.2.1, TAFE 1.4.0 TAFE_Conver1.0.0, TAFE_Conver1.1.0
Customer:	Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0)
End Customer:	ÖBB
SIL Level:	SIL 4 ⁵

Table 2: Security / Maturity Level

Security Level:	SL-T 3, SL-C 3
Maturity Level:	ML 3

⁵ On SIL please refer also to Table 3: WBS managed as Solution

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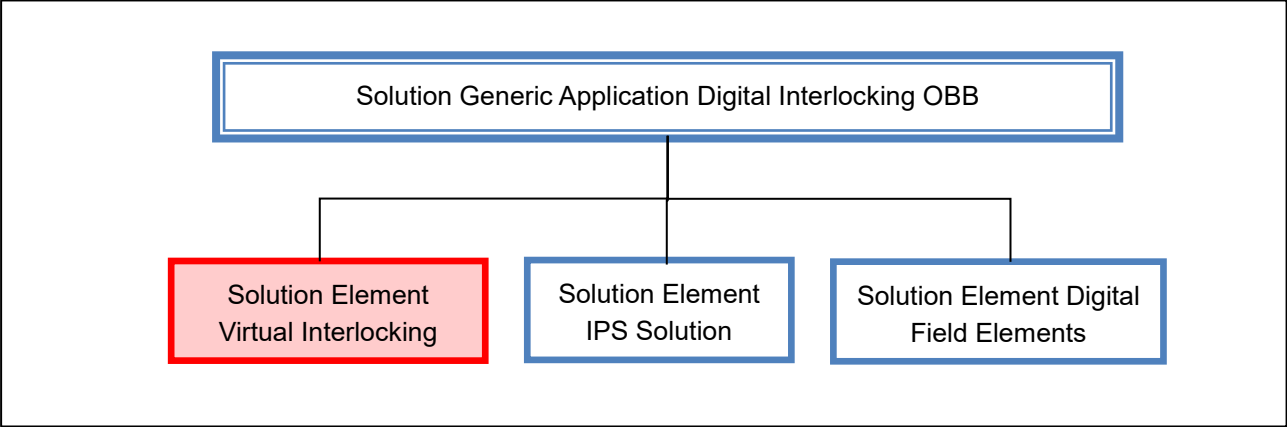
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1.1.2 SOLUTION ELEMENT OF SOLUTION

Figure 1: Solution Overview Generic Application Digital Interlocking OBB (DSTW)



This Solution Element Virtual Interlocking OBB Phase 1 (VIL OBB 1.0.0) marked in red in Figure 1 is a subsystem of the Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0) described in [SEMP DIL].

The summary purpose and scope of the Solution Element Virtual Interlocking OBB Phase 1 (VIL OBB 1.0.0)

- Development:** Creation of the new subsystem „Virtual Interlocking“ (VIL OBB 1.0.0) consisting of integrated software components ELEKTRA Core interlocking generation (ELEKTRA BL OBB 40.00 ELEKTRA PV GEN 2.20.0, CAR-RED_1.0.0) , ECMUX (EEC SW OBB 04.00) and a updated CAE Data Tool called OREST 19.50 with the minimum technical requirements Phase 1 “Basic virtualization” (B-Vi) based on the ÖBB specifications [CRS] „ÖBB_RA-DSTW-D1_Leistungsbeschreibung“ for the Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0)
- Reduction:** Support the goal of Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0) to decrease the need for facilities/buildings for the interlocking ELEKTRA Core.
- Standardization:** Addressing European standardization efforts by railway operators, known as the "EULYNX initiative. This solution element perform implementation along EULYNX BL4R4
- Cyber Security:** Implementation of cyber security measures based on IEC 62443, TS50701, upcoming IEC 63452 and EU Rail System Pillar Cybersecurity Specification V1.0, addressing the NIS2 directive and the European Cyber Resilience Act.
- Integration:** integration of software components ELEKTRA PV GEN 2.20.0, CAR-RED_1.0.0, EEC SW OBB 04.00, OREST 19.50, ELEKTRA BL OBB 40.00 and integration of supplied software from other solutions, TAS Platform Core 2.6.x, TAS Platform OCS 3.6.x, TAS Platform MNT 3.6.x, TAS Platform J4S 3.6.0, RJC RASTA 2.1.0, RJC Core 1.2.0, ELEKTRA OBB 29.50, GNU Chill 2.3.13 and lab integration of DCA OCs.
- Verification:** Test automation and verification of subsystem "Virtual Interlocking" on various levels (module, subsystem, solution element) in the context of Virtual Interlocking according to the IVP and TP.

For a detailed overview about the purpose and scope of the overall solution Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0) see [SEMP DIL].

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The solution Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0) is according CENELEC 50129 responsible for the integration of the subsystem / Solution Element Virtual Interlocking OBB Phase 1 (VIL OBB 1.0.0) intended for use in digital interlocking installations on Austrian railway lines.

The Generic Application will provide requirements and top-level architecture on the system level to the Solution Element and integrate, test, verify, validate and assess the subsystem for further specific application in different Austrian operative projects. Dependencies to planning document of the solution are described in chapter [1.3.3]. For Interaction of the solution and the solution elements see chapter [5.1.1]

1.1.3 WBS MANAGED AS SOLUTION ELEMENT

Table 3: WBS managed as Solution Element

WBS	Identification	SEM	SIL	Initial Product Version	Target Product Version	Target date	Target SEMP
100025795 / ATRELDE91- E0.WP3.DEV. 3.2.1	Solution Element Virtual Interlocking Phase 1 Management	P. Bargiel	SIL4	-	VIL OBB 1.0.0	Milestone FQR, see section 1.2.4	This document
ATREND835- E1.WP3.DEV. 3.2/- E1.WP3.DEV. 3.4	ELEKTRA BASE Phase 0 & Phase 1 Management	P. Bargiel	SIL4	-	VIL OBB 1.0.0	Milestone FQR, see section 1.2.4	This document

Note: This SEMP describes the Basic virtualization" (B-Vi) is the minimum technical requirement to be implemented upon the first delivery of a DSTW; to fulfill the complete set of customer requirements additional product versions are planned and will be described in future versions of the SEMP.

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1.1.4 WBS MANAGED AS WORK PACKAGES

Table 4: WBS managed as Work Packages

WBS	Ident.	WPM	Initial component Version	Target component Version	Target date	Work Package Description
100020889 / ATRENDDB35-E1 ⁶	Work Package ELEKTRA Base Phase 0 & Phase 1 - Component GENSYS_2.20.0	B. Koenig	ELEKTRA PV GEN 2.19.0	ELEKTRA PV GEN 2.20.0	Milestone TQR1, see section [1.3.1]	[WPD EL BASE GENSYS&OBB]
100020889 / ATRENDDB35-E1.WP3.DEV.3.2	Work Package ELEKTRA Base Phase 0 & Phase 1 - Component CAR-RED_1.0.0	P. Bargiel	CAR_5.1.2	CAR-RED_1.0.0	Milestone TQR1, see section [1.3.1]	[WPD EL BASE CAR-RED]
100025795 / ATRELDE91-WP3.DEV.3.2.1.2.SW	Work Package ELEKTRA ÖBB Phase 1	B. Koenig	ELEKTRA BL OBB 29.21	ELEKTRA BL OBB 40.00	Milestone TQR1, see section [1.3.1]	[WPD EL BASE GENSYS&OBB]
100025795 / ATRELDE91-WP3.DEV.3.2.1.3	Work Package ECMUX Phase 1	R. Rambo	-	EEC SW OBB 04.00	Milestone TQR1, see section [1.3.1]	[WPD ECMUX]
100025795 / ATRELDE91-WP3.DEV.3.2.1.4	Work Package CAE Data Tools Phase 1	R. Ungerboeck	OREST 19.20	OREST 19.50	Milestone TQR1, see section [1.3.1]	[WPD OREST]
100025795 / ATRELDE91-WP3.DEV.3.2.1.5	Work Package ELEKTRA Test Automation Phase 0 & Phase 1	A. Chita	-	TAFE 1.4.0, TAFE_Converter_1.1.0	Milestone TQR1, see section [1.3.1]	[WPD Test Autom]

⁶ Consisting of several WBS IDs a separate Project ID. The detailed WBS ID is described in [WPD EL BASE GENSYS&OBB]

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ATRENDDB35- E1.WP3.DEV.T AFE	Work Package Test Automation Framework Phase 1	E. Diethelm	TAFE 1.0.0	TAFE 1.4.0	Milestone TQR1, see section [1.3.1]	[WPD TAFE 1.2.1]
100020889 / ATRENDDB35- E1.WP3.DEV.T AFEC	Work Package Test Automation Framework Convertor	E. Diethelm	-	TAFE_Con ver1.1.0	Milestone TQR1, see section [1.3.1]	[WPD TAFE Convertor_1.0.0]
100025795 / ATRELDE91- E0.WP3.DEV.3. 2.1.3.SW	Work Package VIL Testing Phase 1	D. Olah	-	-	Milestone TQR1, see section [1.3.1]	[WPD VIL Testing]

A detailed view on all WBS diagram is listed in section [3.6].

1.1.5 WBS MAPPING TO PROJECT IDS

This Solution Element Virtual Interlocking OBB Phase 1 (VIL OBB 1.0.0) consists of two project IDs. Several Work packages and management activities are managed and reported under a separate Project ID. A mapping can be found in the following Table 5: WBS mapping to Project IDs:

Table 5: WBS mapping to Project IDs

Project ID	Work packages
100025795 / ATRELDE91	- Solution Element Virtual Interlocking Phase 1 Management - Work Package ECMUX Phase 1 - Work Package CAE Data Tools Phase 1 - Work Package VIL Testing Phase 1
100020889 / ATRENDDB35	- ELEKTRA BASE Phase 0 & Phase 1 Management - Work Package ELEKTRA Base Phase 0 & Phase 1 - Component GENSYS_2.20.0 - Work Package ELEKTRA Base Phase 0 & Phase 1 - Component CAR-RED_1.0.0 - Work Package ELEKTRA ÖBB Phase 1 - Work Package ELEKTRA Test Automation Phase 0 & Phase 1 - Work Package Test Automation Framework Phase 1 - Work Package Test Automation Framework Convertor

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1.2 SUBSYSTEM OVERVIEW

1.2.1 SUBSYSTEM OF SYSTEM

In the context of architecture, the Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0), is also referred as system.

In the context of architecture, the Solution Element Virtual Interlocking OBB Phase 1 (VIL OBB 1.0.0) is also referred as subsystem

In the context of architecture, the

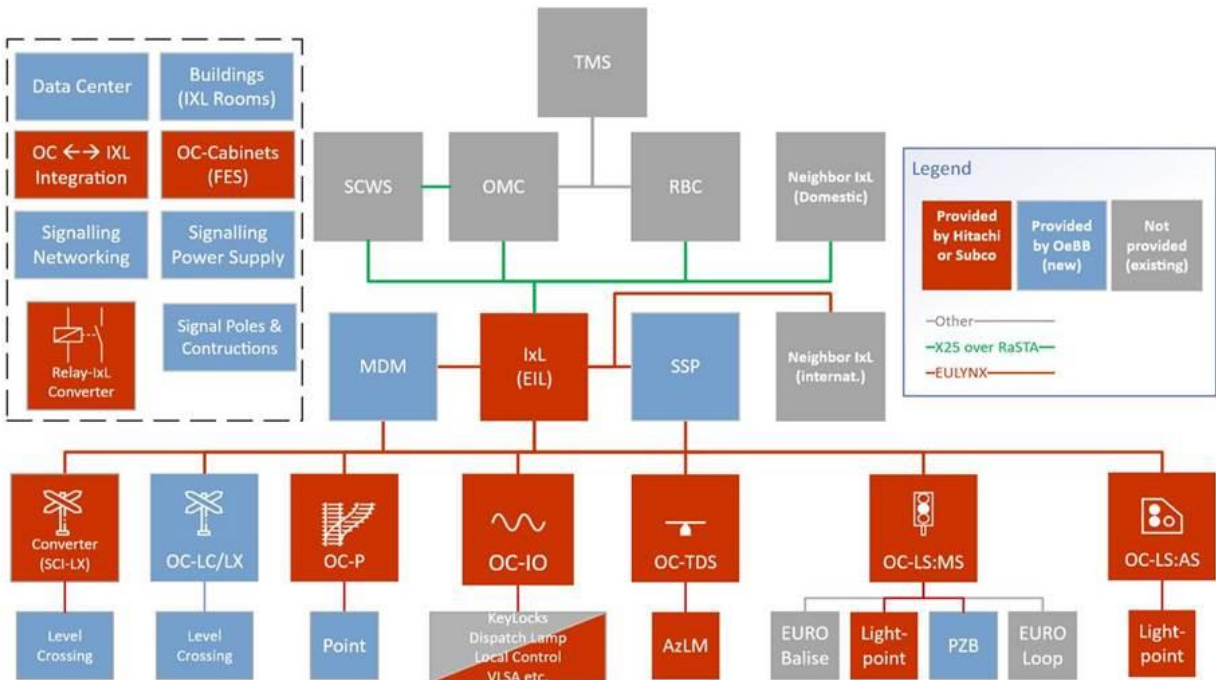
- Work Package ECMUX Phase 1
 - Work Package ELEKTRA Base Phase 0 & Phase 1 - Component GENSYS_2.20.0
 - Work Package ELEKTRA ÖBB Phase 1
- are also referred as software components.

In the context of architecture Work Package ELEKTRA Base Phase 0 & Phase 1 - Component CAR-RED_1.0.0 are also referred as subcomponent or element.

In the context of architecture Features e.g. DGP, CC, RCU, Keylock, Point, etc. of the work packages are referred as elements.

1.2.2 SYSTEM SCOPE SPLIT

For an overview about customer expectation see: [CRS] „OBB_RA-DSTW-D1_Leistungsbeschreibung “.



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Figure 2: Scope Split below gives an overview of the complete Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0) and how the Solution Element Virtual Interlocking OBB Phase 1 (VIL OBB 1.0.0) is embedded. Solution Element Virtual Interlocking OBB Phase 1 (VIL OBB 1.0.0) and Solution Element IPS Solution OBB 1.0.0 (IPSS OBB 1.0.0) are subsystems embedded in Electronic Interlocking (EIL) also name as DIL-CP marked in red. Electronic Interlocking (EIL) shall be designed as a computer cluster that performs the central interlocking control function for multiple interlocking areas (stations).

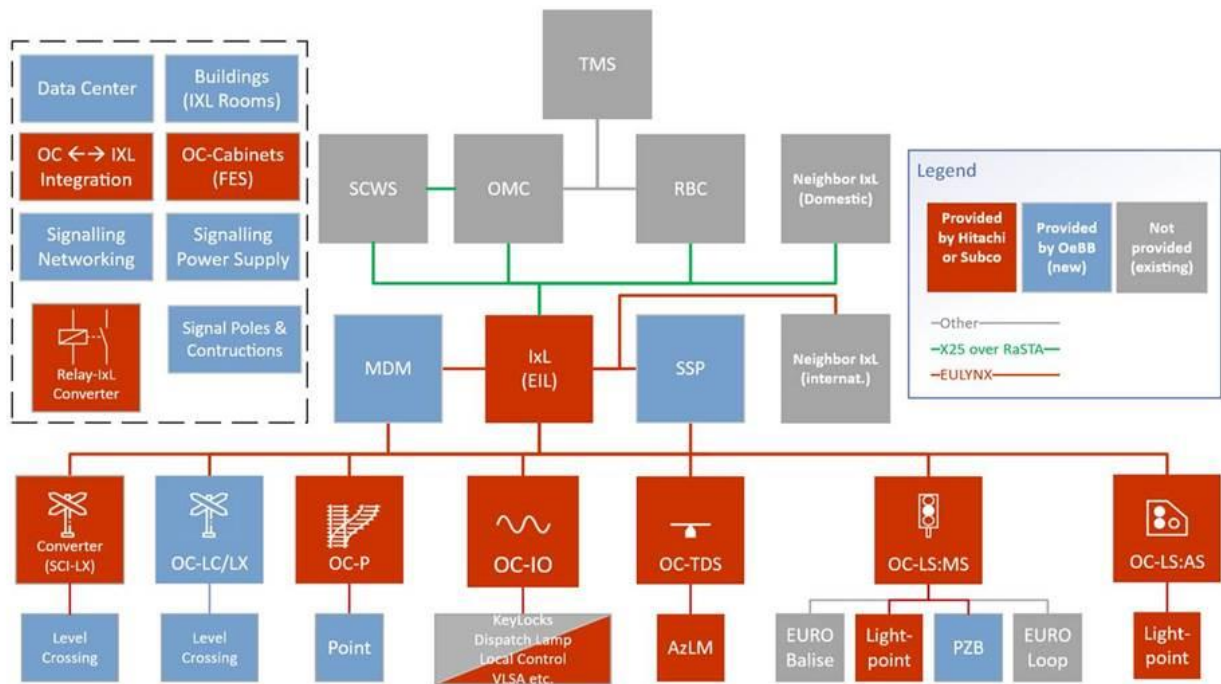


Figure 2: Scope Split

For more information about used products see [PBS].

1.2.3 SUBSYSTEM ARCHITECTURE

The following Figure 3: Digital Interlocking System Architecture is an excerpt of [SysArch] illustrates the top-level system architecture. In the Electronic Interlocking (EIL) the Solution Element Virtual Interlocking OBB Phase 1 (VIL OBB 1.0.0) marked in red illustrates the subsystem. It uses the surrounding central virtualization platform including necessary administration workflows delivered by Solution Element IPS Solution OBB 1.0.0 (IPSS OBB 1.0.0).

For a more comprehensive overview to the Subsystem and System Solution Element Virtual Interlocking OBB Phase 1 (VIL OBB 1.0.0) architecture please refer to [SSDD VIL] & [SSDD DIL].

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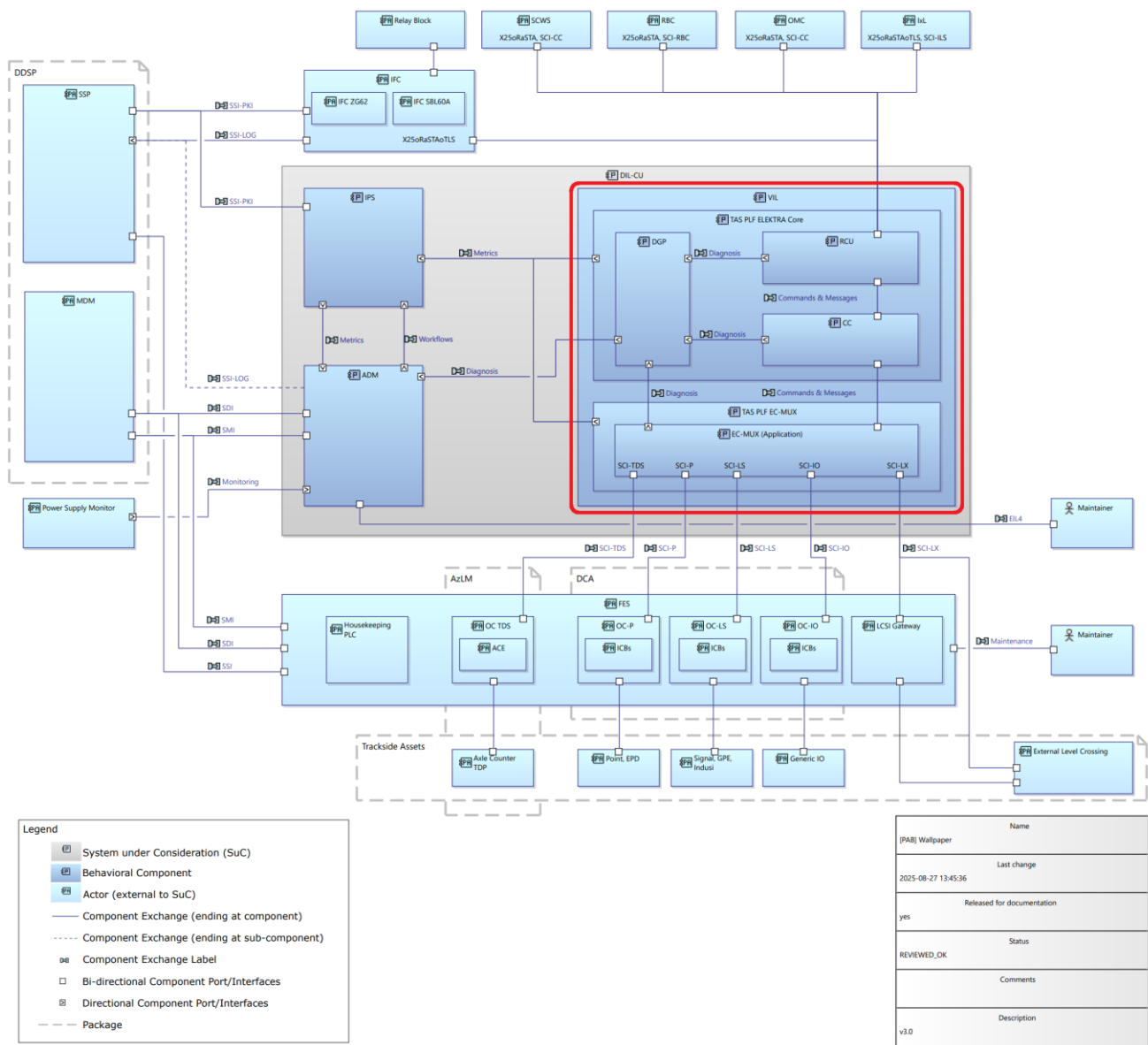


Figure 3: Digital Interlocking System Architecture from [SysArch]

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1.2.4 SUBSYSTEM DESIGN, DEVELOPMENT & INTEGRATION

The design, development and lab integration of subsystem software components of Solution Element Virtual Interlocking OBB Phase 1 (VIL OBB 1.0.0) are managed as Work packages described in more detail in chapter [1.1.4]. Figure 4 shows a brief visual description about the configuration of the components. All components focuses on the Phase 1 minimum technical requirement “Basic Virtualization” (B-Vi), based on ÖBB specifications [CRS] and the requirements from [SSS DIL], [SSS SEC DIL], [SSS VIL], [SSS SEC VIL]. Some components also have Phase 0 features in scope:

- Work Package ELEKTRA Base Phase 0 & Phase 1 - Component GENSYS_2.20.0 described in [WPD EL BASE GENSYS&OBB] covers the development of the generic software part of the TAS PLF ELEKTRA Core illustrated in Figure 3: Digital Interlocking System Architecture. This work package also includes the integration of subcomponent CAR-RED_1.0.0.
- Work Package ELEKTRA Base Phase 0 & Phase 1 - Component CAR-RED_1.0.0 described in [WPD EL BASE CAR-RED] covers the development of the elements CAR library, EP handler library, & CAR console. These elements support the CHILL Application Runtime (CAR) Redundant (RED), which runs on a redundant TAS Platform ELEKTRA Core setup in hot-hot mode and is situated in communication, messages & diagnosis between RCU, CC, DGP illustrated in Figure 3: Digital Interlocking System Architecture.
- Work Package ELEKTRA ÖBB Phase 1 described in [WPD EL BASE GENSYS&OBB] covers the development of the country specific software part of the TAS PLF ELEKTRA Core illustrated in Figure 3: Digital Interlocking System Architecture. This work package also includes the integration of Component ELEKTRA PV GEN 2.20.0.
- The work package Work Package ECMUX Phase 1 described in [WPD ECMUX] covers SW development for component TAS Plf. ECMUX & ECMUX Application illustrated in Figure 3: Digital Interlocking System Architecture and contains the development of elements IO DispatchLamp, IO Keylocks, IO UnivElem, IO Generic, ÖBB Light Signals, TDS AzLm 7.10, Power Supply, ECMUX on Server (2x1001), Diagnose & Maintenance, supporting Security Phase 1 and lab integration of OC-IO OC-P Signals (OC-LS), TDS AZLM (OC-TDS) and io elements (OC-IO).

1.2.5 SUBSYSTEM CAE DATA PREPARATION TOOLS

For the Subsystem mentioned in section [1.2.4] the CAE Data Preparation Tool “OREST” is enhanced in Work Package CAE Data Tools Phase 1 described in [WPD OREST].

1.2.6 SUBSYSTEM VERIFICATION

The Subsystem mentioned in section [1.2.4] a verification on the subsystem level is performed by Work Package VIL Testing Phase 1 described in [WPD VIL Testing] and supported by subsystem test automation Work Package ELEKTRA Test Automation Phase 0 & Phase 1 described in [WPD Test Autom] with the support of enabling tools Work Package Test Automation Framework Convertor described in [WPD TAFE Convertor_1.0.0] & Work Package Test Automation Framework Phase 1 described in [WPD TAFE 1.2.1]

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1.2.7 SUBSYSTEM VALIDATION

The Validation of the Subsystem is managed the ELEKTRA BASE Phase 0 & Phase 1 Management, Solution Element Virtual Interlocking Phase 1 Management and this document. Deliverables are mentioned in chapter [1.3.2].

1.3 KEY POINTS

1.3.1 MILESTONES FOR SOLUTION ELEMENT

Table 6: Milestones Solution Element Virtual Interlocking OBB Phase 1 (VIL OBB 1.0.0)

Milestone	Target Date	Lead
Solution Element Orientation Review 1 (SOR1)	04.04.2025	SEM
Solution Element Orientation Review 2 (SOR2)	T0 ⁷ + 1 month (expected 01/26)	SEM
Solution Element Requirement Review (SRR1)	07.10.2025	SEM
Solution Element Requirement Review (SRR2) ⁸	MS E03 ⁹ & Input Data see Footnote ⁸ + 3 month (expected 07/26)	SEM
Solution Element Functional Review (SFR) ¹⁰	MS E03 ⁹ & Input Data see Footnote ⁸ + 5 month (expected 09/26)	SEM
Solution Element Preliminary Design Review (PDR)	Agile approach ¹¹	SEM

⁷ All dates are under the assumption of a contract signature by end of December 2025 (=T0). For latest available dates see [IMP].

⁸ Input data [SSS DIL], [SSS SEC DIL], [SecTRA], [MCRD DIL], [MCRD OBB], [SSDD DIL] released. More details on input data delivery date see chapter [3.4]. For Interaction of requirements solution and solution elements see [RMP] & process see chapter [5.1.1].

⁹ MS E03 Pflichtenheft Freigabe from Customer OBB defined in [PMP]. MS E03 is currently forecasted with April 2026. For latest available dates see [IMP].

¹⁰ Precondition is [SSDD DIL] & completion of all EPICs with definition of Specification Done according to [S32] see chapter [Error! Reference source not found.Template of DoX Templates]. Input data is described in chapter [3.4]. For interaction of solution and solution elements see chapter [5.1.1]. After completion of all Definition of Specification Done following template [S32] & Agile Definition [S25]. More comprehensive overview see chapter [Error! Reference source not found.Template of DoX Templates].

¹¹ Iterative agile approach. Visibility Points in design and implementation with deliverables motioned in chapter [1.3.2]. Iterative check on EPICs within the Work packages following the Agile Definition [S25] & Definition of EPIC Done Template following [S32]. More comprehensive overview see chapter [5.2.2] & [5.2.3Template of DoX Templates].

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Solution Element Critical Design Review (CDR)	Agile approach ¹¹	SEM
Test Readiness Review - WP ELEKTRA Test Automation (TRR - WP TA)	01.11.2026	WPM ELEKTRA Test Automation
Test Readiness Review - WP VIL ECMUX Phase 1 (TRR – ECMUX)	01.12.2026	WPM ECMUX
Technical Qualification Review 1 (TQR1-WP CAE Data Tools) OREST 19.60	26.03.2027	WPM CAE Data Tools
Test Readiness Review 1 - WP ELEKTRA BASE Phase 1 GENSYS 2.20.0 & OBB 40.00 (TRR – WP EL BASE)	01.08.2027	WPM ELEKTRA BASE
Solution Element & WP VIL Testing Test Readiness Review 1 (TRR1)	01.08.2027	IVM
Solution Element Technical Qualification Review 1 (TQR1) ¹²	01.12.2027	IVM
Solution Element Test Readiness Review 2 (TRR2)	01.12.2027	IVM
Solution Element Technical Qualification Review 2 (TQR2)	01.05.2028 ¹³	SEM
Solution Element Final Qualification Review (FQR)	01.06.2028	SEM

1.3.2 SOLUTION ELEMENT DELIVERABLES

Table 7: Deliverables of Solution Element Virtual Interlocking OBB Phase 1 (VIL OBB 1.0.0)

ID	Description	Target Date	Customer
VIL_VP1	Showcase of developer version OBB_40.00 "OBB_EL4_VP1" containing GENSYS 2.20.0 "GEN_CCX_EL4_VP1" CCA/CCB on redundant NMR	23.04.2024	Generic Application

¹² The Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0) integrate, test, verify, validate and assess the solution element for further specific application is performed in a different Austrian operative project see [PMP]. For Interaction of solution and solution elements see chapter [5.1.1]

¹³ This milestone is performed together Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0) DIL TQR2. For Interaction of solution and solution elements see chapter [5.1.1]

Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0)

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	1x2oo2 together with integration baseline "CAR-RED_EL4-VP1". see[WPD EL BASE GENSYS&OBB]		Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0)
VIL_VP2	Showcase of developer version OBB_40.00 "OBB_EL4_VP2" containing GENSYS 2.20.0 "GEN_CCX_EL4_VP2" CCA and CCB applications can run as 2x2oo2 (=hot-hot redundant) with integration baseline "CAR-RED_EL4-VP2".see [WPD EL BASE GENSYS&OBB] & [WPD EL BASE CAR-RED]	01.07.2024	Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0)
VIL_VP3	Showcase of developer version OBB_40.00 "OBB_EL4_VP3" containing integrated applications of GENSYS 2.20.0 (CC, RCU, DGP) running on 2x2oo2 redundant system see [WPD EL BASE GENSYS&OBB] & intermediate overview about CAR-RED_EL4-VP2 testing activities. see [WPD EL BASE CAR-RED]	07.11.2024	Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0)
VIL_VP4	1) Showcase of developer version OBB_40.00 "OBB_EL4_VP4" containing integrated GENSYS 2.20.0 & CAR-RED 1.0.0 applications start-up redundant with feature ECMUX on Server Prototype element point. see [WPD EL BASE GENSYS&OBB] 2) Test execution performed and test report provided for dev. version "OBB_EL4_VP3" Details see [WPD VIL Testing]	05.03.2025	Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0)
VIL_VP5	Review [CMR Ed.03] deliverables: 1) GEN_OBB_VP5_D1, GEN_OBB_VP5_D2 see [WPD EL BASE GENSYS&OBB] 2) CARRED_VP5_D01, CARRED_VP5_D02, CARRED_VP5_D03, CARRED_VP5_D04 see [WPD EL BASE CAR-RED] 3) TA_VP5_D01 see [WPD Test Autom] 4) VILTST_VP5_D01, VILTST_VP5_D02, VILTST_VP5_D03 see [WPD VIL Testing]	15.07.2025	Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0)
VIL_VP6	1) Showcase of developer version OBB_40.00 "OBB_EL4_VP4" with integrated GENSYS 2.20.0 developments GEN_OBB_VP6_D1 Real Time Clock synchronization with NTP & GEN_OBB_VP6_D2 X25overRASTA on RCA/RCB and integrated CAR-RED_EL4-VP5 baseline. see [WPD EL BASE GENSYS&OBB] 2) Showcase of developer version EEC_SW_OBB_04.00 with TDS AZLM 7.9 see [WPD ECMUX] 3) Review [CMR Ed.03] deliverables	24.09.2025	Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0)

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	a) GEN_OBB_VP6_D1, GEN_OBB_VP6_D2 GEN_OBB_VP6_D3 see [WPD EL BASE GENSYS&OBB] b) CARRED_VP6_D01, CARRED_VP6_D02 see [WPD EL BASE CAR-RED] c) TA_VP6_D03 see [WPD Test Autom] d) VILTST_VP6_D01, VILTST_VP6_D02, VILTST_VP6_D03 see [WPD VIL Testing]		
VIL_VP7	1) Showcase of developer version EEC_SW_OBB_04.00 with Dispatch Lam / Univelement integrated with real DCA 3 deliverable, Eulynx BL2 (Neupro) IO Module see [WPD ECMUX] 2) Review of deliverables [CMR Ed.03] a) GEN_OBB_VP7_D1, GEN_OBB_VP7_D2, GEN_OBB_VP7_D3, GEN_OBB_VP7_D4, see [WPD EL BASE GENSYS&OBB] b) CARRED_VP7_D01, CARRED_VP7_D02, CARRED_VP7_D03, CARRED_VP7_D04 see [WPD EL BASE CAR-RED] c) VILTST_VP7_D01, VILTST_VP7_D02, VILTST_VP7_D03, VILTST_VP7_D04, VILTST_VP7_D05 see [WPD VIL Testing]	15.12.2025	Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0)
VIL_D1	1) Release of this document 2) TBA	Milestone SOR2, see section 1.2.4	Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0)
VIL_VP8	1) Showcase of developer version OBB_40.00 "OBB_EL4_VP8" with integrated a) GENSYS 2.20.0 developments CMOS-Data, DGP, Rework DGP, b) Plf 2.6.0 & Plf 2.6.1 c) EEC_SW_OBB_04.00 ECMUX on Server with developer version EULYNX BL 4.2 OBB Keylocks Line and Station based on EULYNX generic IO on EULYNX BL 4.2 see [WPD ECMUX] 2) Review of deliverables [CMR Ed.03] a) GEN_OBB_VP8_D1, GEN_OBB_VP8_D2, GEN_OBB_VP8_D3, GEN_OBB_VP8_D4, GEN_OBB_VP8_D5, GEN_OBB_VP8_D6, GEN_OBB_VP8_D7 see [WPD EL BASE GENSYS&OBB] b) TA_VP8_D01, TA_VP8_D02 see [WPD Test Autom]	14.04.2026	Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0)

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	c) VILTST_VP8_D01, VILTST_VP8_D02, VILTST_VP8_D03, VILTST_VP8_D04, VILTST_VP8_D05 see [WPD VIL Testing]		
VIL_D2	1) Release of document set Subsystem Specifications [SSS VIL] & [SSS SEC VIL]. 2) Release of document set Verification Cross Reference Matrix, see [DoP] 3) Release of document set Validation of Requirements Report, ANV see [DoP] 4) Release of [QMP VIL] 5) Release of [RAMP VIL]	Milestone SRR2, see section 1.2.4	Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0)
VIL_D3	Release of Subsystem Virtual Interlocking Architectural Specification [SSDD VIL].	Milestone SFR, see section 1.2.4	Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0)
VIL_VPx	Future Developer Versions with Points – SWCRR DCA3 Signals – SWCRR TDS – SWCRR SWCRR	TBA	Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0)
VIL_D4	Release of document set Safety Concepts, see [DoP]	TBA	Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0)
VIL_D4	Data preparation rules / directives based on developer version OREST 19.50	Milestone TRR1 CAE Data Tools, see section 1.2.4	Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0)
VIL_D5	Customer Acceptance Test, see [IVP]	TBA	OBB

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VIL_D5	1) ClearCase views for customer data preparation containing delivered baseline and associated tools of the new integrated software solution ELEKTRA Core interlocking generation („Virtual Interlocking) ELEKTRA BL OBB 40.00 with the minimum technical requirement Phase 1 “Basic virtualization” (B-Vi) based on the ÖBB specifications [CRS] „OBB_RA-DSTW-D1_Leistungsbeschreibung“ for the Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0) with integrated sw components ELEKTRA PV GEN 2.20.0, CAR-RED_1.0.0, EEC SW OBB 04.00 2) Release of Final Data preparation rules / directives, see [DoP] 3) Release of BL Delivery Sheet Solution Element Virtual Interlocking OBB Phase 1 (VIL OBB 1.0.0), see [DoP] 4) Release of Tool Classification EN50128 / EN50129, see [DoP]	Milestone TQR1, see section 1.2.4	Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0)
VIL_D6	1) Release of document set Safety Cases, see [DoP] 2) Release of document set Security Cases, see [DoP] 3) Release of document set Security Evidences of Compliance, see [DoP] 4) Release of document set Security Guidelines, see [DoP] 5) Release of document set Security Vulnerability Management Reports, see [DoP]	Milestone TQR2, see section 1.2.4	Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0)
VIL_D7	1) Release of [DoP] 2) Release of Fault lists 3) Lessons Learnt Report Virtual Interlocking	Milestone FQR, see section 1.2.4	Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0)

Following items listed under below are planned to be implemented till a stated target phase and are not part of this product version delivery. They will be developed, tested and released in a successor product version.

Short	Description	Phase to be finished optionally

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Level Crossing (using Eulynx)	Design of a set of functions for Eulynx Level Crossing external interface to object controller to external supplier, see [WPD ECMUX]	Milestone TQR1, see section 1.2.4
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1.3.3 WORK PACKAGE DELIVERABLES

Additional deliverables of the component level handled as work packages of this Solution Element please refer to Work package descriptions in mentioned in chapter [1.1.4].

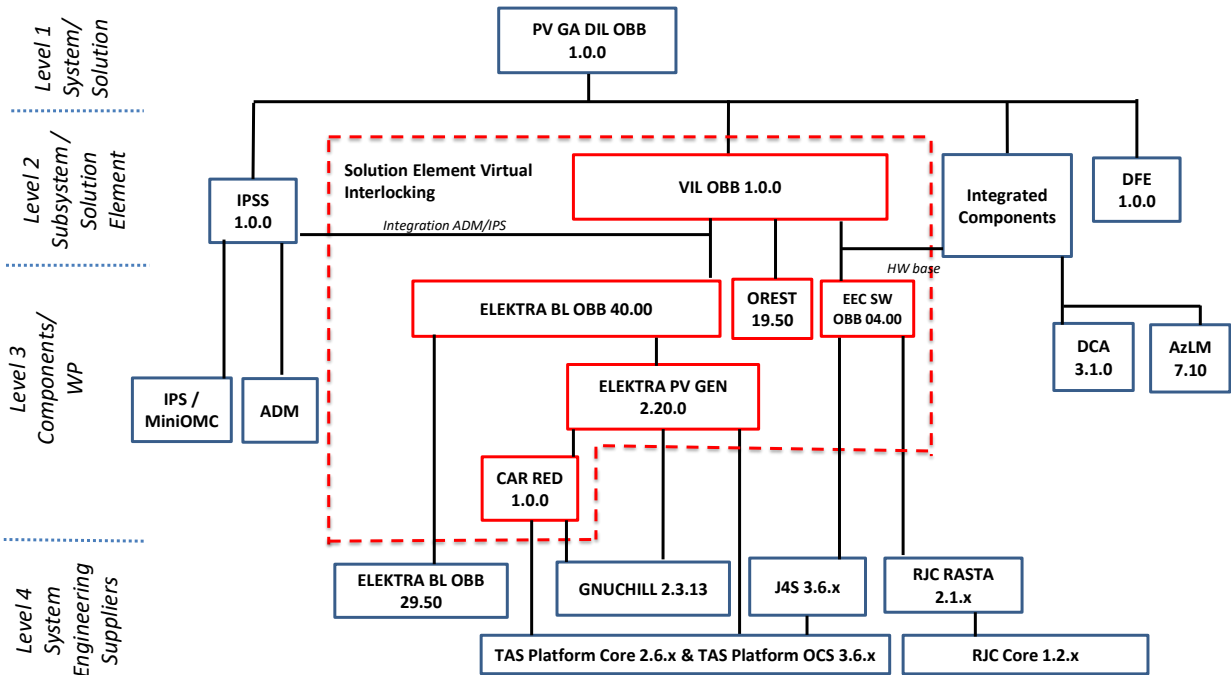
1.4 BASELINE OF SOFTWARE CONFIGURATION ITEMS AND DOCUMENTATION LABELS

1.4.1 SUBSYSTEM CONFIGURATION

The subsystem configuration is managed by the solution element.

The initial and target versions of this solution element and their work packages for this engineering project are listed in Table 3: WBS managed as Solution. Table 4: WBS managed as Work Packages gives an overview about the versions of this solution element:

Figure 4: Overview about the versions of this solution element



For more information about used products see [PBS] or details on the Product versions see [CMP]. Initial version see section [1.1.4].

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1.4.2 DOCUMENTATION LABELS

For documentation available via DWF following labels are used as defined also in the Documentation Plan [DoP]. The labelling shall follow the definition of Target Product Versions as per Table 3: WBS managed as Solution.

Details on the Product versions are given in the [CMP].

For Documents on the Solution Element Virtual Interlocking Phase 1 Level following labels shall be applied

Solution Element/ Component	Target Version	Document Label	ANV Range	Responsible for DOC-NO Allocation
Subsystem Virtual Interlocking	VIL OBB 1.0.0	VIL_OBB_1.0.0	3BU 15000 6950 - 6989	DM, see section [Error! R eference source not found.]
Virtual Interlocking Generic System Product Version	ELEKTRA PV GEN 2.20.0	PV_EL_GEN_2.20.0	3BU 15000 6850 - 6899	
Virtual Interlocking ÖBB System Product Baseline	ELEKTRA BL OBB 40.00	BL_OBB_40.00	3BU 15000 6900 - 6949	
Virtual Interlocking Chill Application Runtime redundant	CAR REDUNDANT 1.0.0	CAR-RED_1.0.0	3BU 15000 6800 - 6849	
Virtual Interlocking SEC/ECMUX within the DSTW project	SEC SW OBB 04.00	SEC_SW_OBB_04.00	3BU 61400 0000 - 9999	ARC WP ECMUX see section [Error! R eference source not found.]
CAE Data Preparation Tools within the DSTW project	OREST 19.30 OREST 19.40 OREST 19.50	OREST_19.30 OREST_19.40 OREST_19.50	3BU 15000 6400 - 6799	Sascha Krasny
Work Package Test Automation Framework Phase 1	TAFE 1.4.0	TAFE_1.2.1 TAFE_1.3.0 TAFE_1.4.0	3BU 15000 4377 - 4379	Eric DIETHELM
Virtual Interlocking Test Automation Framework Convertor for Virtual Interlocking Test Automation Work package	TAFE Converter 1.1.0	TAFE_Converter_1.0.0 TAFE_Converter_1.1.0	3BU 00605 0000 - 9999	Eric DIETHELM

1.5 DOCUMENT VERSION HISTORY

1.5.1 EDITION 01

Initial edition

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1.5.2 OPEN POINTS

The following open points will be addressed in the next edition:

- Section WBS managed as Work Packages 1.1.4, Documentation Labels 1.4.2 – Fix target version TAFE, Tafe convetor relevant TAFE Versions for SE VIL
- Section Solution Element Deliverables 1.3.2 – Finalize deliverables and proposed date for future Visibility Points and align on proposed dates of some deliverables e.g. Customer Acceptance Test, Safety concepts, Verification report, validation reports, etc.
- Section Referenced documents 2.1 – Update of references when ANV avail. In DoPs
- Section Terms, abbreviations and acronyms 2.4.- Reference to glossary term imported and released in glossary.
- Section System engineering supplies 3.2 - Target Versions of SYSTEM ENGINEERING SUPPLIES Solutions.
- Section Tools and Facilities 3.12 - Complete input from ARCs, WPMs and Solution Core Team Tools and facilities.
- Section Solution element core team 3.1.1, Solution element core Agile team 3.1.2, 3.3.1 – Update NN when defined. Update where TBA.
- Complete Input data 3.3 where stated TBA.
- Section 5.6 Tailoring of the monitoring of project progress within an agile development environment – Complete content
- Section Preconditions of solution Element 3.4.1 VIL_P08
- Section 5 - Description of the tailoring in to Work Package CAE Data Tools Phase 1, Work Package ELEKTRA Test Automation Phase 0 & Phase 1, Work Package Test Automation Framework Phase 1, Work Package Test Automation Framework Convertor
- Section 3.2. Schutzweg Feature TBD which target OBB Baseline.
- Section 5.2.5 Security certification for Subsystem handling, deliverables not defined yet.
- Section 5.1.1 Software/Code Verification = VER?
- Section appendix A does not list if the tables are from En5128 or from the newer EN50716. this should be checked, it should be according En50716.

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2. REFERENCES AND TERMS

2.1 REFERENCED DOCUMENTS

The relevant version of a document is referenced in the Documentation Plan [DoP]. Several planning documents are concurrent in progress until milestone SOR, therefore the edition 01 is referenced of Documentation Plan / Process Verification Report. Until milestone FQR the Documentation Plan / Process Verification Report for this Solution will be updated and tracked within the Version in TOL see TRN [DoP].

Table 8 : Referenced Documents

Id	Nr./Title	Reference	Version
[DoP]	Documentation Plan (DoP)	<u>3BU 15000 6950 ADAPA</u> <u>DoP_VIL.xlsm</u> <u>3BU 15000 6950 ADAPA</u>	latest 01PD01

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Id	Nr./Title	Reference	Version
[CRS]	DSTW Streckenausrüstung	3BU 17300 0004 DSAPC	04
	OBG - Teil D	3BU 17300 0050 DTAPC	01PD01
	Leistungsbeschreibung und Lastenhefte	3BU 17300 0051 DTAPC	01 (plan.)
	Pflichtenheft Integrationsplan	3BU 17300 0030 DTAPC	02
	Pflichtenheft - Security-Konzept	3BU 17300 0031 DTAPC	02
	Pflichtenheft Digital	3BU 17300 0023 DTAPC	02
	Interlocking OBB - D2.10	3BU 17300 0024 DTAPC	02
	Teilpflichtenheft Teilsystem Gleisfreimeldung	3BU 17300 0025 DTAPC	01PD02
	Pflichtenheft Digital	3BU 17300 0026 DTAPC	02
	Interlocking OBB - D2.11	3BU 17300 0027 DTAPC	02
	Teilpflichtenheft Field Element Station	3BU 17300 0028 DTAPC	02
	Pflichtenheft Digital	3BU 17300 0029 DTAPC	02
	Interlocking OBB - D2.2		
	Teilpflichtenheft RAMS		
	Pflichtenheft Digital		
	Interlocking OBB - D2.3		
	Teilpflichtenheft Allgemeine Anforderungen		
	Pflichtenheft Digital		
	Interlocking OBB - D2.4		
	Teilpflichtenheft Security		
	Pflichtenheft Digital		
	Interlocking OBB - D2.5		
	Teilpflichtenheft Teilsystem DSTW-ZE		
	Pflichtenheft Digital		
	Interlocking OBB - D2.6		
	Teilpflichtenheft Teilsystem Weiche		
	Pflichtenheft Digital		
	Interlocking OBB - D2.7		
	Teilpflichtenheft Teilsystem Lichtsignal		
	Pflichtenheft Digital		
	Interlocking OBB - D2.8		
	Teilpflichtenheft Externe EKSA		
	Pflichtenheft Digital		
	Interlocking OBB - D2.9		
	Teilpflichtenheft Teilsystem Generische Ein- und Ausgabe		
[PMP]	Project Management Plan	3BU 17300 1000 DPAPC	01PD02

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Id	Nr./Title	Reference	Version
[RiskReport]	Risk Report	100025795_ATRELDE91-E0 DSTW ÖBB Entwicklung_Risk_List_20250930_ED02 .xlsx	02
[EngDB SE VIL]	Engineering Dashboard ATRELDE91_SEVIL	EngDB_ATRELDE91_SEVIL	Latest
[EngDB EL BASE & TA]	Engineering Dashboard ATREND35_WPELBASE_W PTA	EngDB_ATREND35_WPELBASE_WP TA	latest
[EngDB DIL]	Engineering Dashboard ATRELDE91_GADIL	100025795_ATRELDE91_DSTW-ÖBB	latest
[QMP DIL]	Quality Management Plan DSTW	3BU_17300_1202_QMAPC	01PD01
[QMP VIL]	Quality Management Plan ELEKTRA	3BU_15000_5244_QMAPA	08PD01
[PSP]	Safety Plan DSTW	3BU_17300_1204_DUAPA	01
[SoC]	Scope of Change (SoC)	-	n.a. ¹⁴
[DVa]	Design Validation approval (DVa)	OeBB_Bid_DVa.docx 3BU_17300_0010_DSAPC	12 04
[ToolClassSW]	Software Development, Tool Classification, EN 50128	3BU_15000_5135_AAAPA¹⁵	latest
[ToolClass]	Thales Austria Tool Classification EN50129	3BU_15000_5135_190EN	latest
[CMP]	DSTW CMP	3BU_17300_1210_DUAPA	01 planned
[CMP GEN]	Configuration Management Plan ELEKTRA	3BU_15000_3338_QMAPA	11
[RMP]	Requirements Management Plan (RMP) Generic Application Digital Interlocking ÖBB	3BU_17300_1207_DUAPA	01
[RAMP GEN]	RAM Plan Generic	3BU_15000_1712_DUAPA	06

¹⁴ Changes which are monitored over PCRs.

¹⁵ This document will need to be updated according to the successor [S2] EN 50716:2024 – Railway application – Requirements for software development of EN 50128

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Id	Nr./Title	Reference	Version
[RAMP VIL]	RAMP Plan Solution Element Virtual Interlocking	3BU 17300 1209 DUAPA	01
[MS-Report]	Milestone Report Solution Element Virtual Interlocking	3BU 15000 6956 ADAPA MSR_VIL.xlsm 3BU 15000 6956 ADAPA	Latest 01 planned
[OBB_RA-DSTW]	OBB_RA-DSTW-D1_Leistungsbeschreibung	3BU 17300 0004 DSAPC	04
[WBS&OBS]	Work Breakdown Strukturen and Organisational Breakdown Structure	3BU 15000 6955 BCAPA	01
[IVP]	Project Specific Integration and Verification Plan (IVP)	3BU 17300 1206 DUAPC	01
[VQP]	Project Specific Validation and Qualification Plan (VQP)	3BU 17300 1205 DUAPA	01
[SecMP]	Project Specific Security Management Plan (SecMP)	3BU 17300 1213 DUAPA	01PD02
[PBS]	Product Breakdown Structure	3BU 17300 1215 DPAPA	02
[SEMP SE DFE]	SEMP Solution Element - Digital Field Elements	3BU 62800 0000 DPAPA	02
[SEMP SE IPSS]	System Engineering Management Plan Solution Element IPSS	3BU 17300 2001 DPAPA	02
[SEMP DIL]	System Engineering Management Digital Interlocking OBB	3BU 17300 1200 DPAPA	02
[IMP]	Integrated Master Schedule	Eng_Master_Schedule_in_SharePoint	latest
[SSS SEC DIL]	System Segment Specification (SSS) - Cybersecurity DSTW ÖBB	3BU 17300 1252 DTAPA	01 (planned)
[SSS VIL]	Subsystem Segment Specification (SSS) Virtual Interlocking	3BU 15000 6955 DTAPA	01PD05
[SSS SEC VIL]	Subsystem Segment Specification (SSS) - Cybersecurity Virtual Interlocking	3BU 15000 6980 DTAPA	01PD04

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Id	Nr./Title	Reference	Version
[SSS DIL]	System Segment Specification (SSS) - Generic Application - Digital Interlocking OBB	3BU_17300_1251_DTAPA	01PD03
[MCRD DIL]	Feature List DSTW	3BU_17300_0000_FLAPC	01 (planned)
[MCRD OBB]	Feature List Elektra ÖBB	3BU_15000_4301_FLAPC	42
[SSDD DIL]	SSDD Digitales Stellwerk - Zentraleinheit	3BU_15000_1254_DSAPA	01 PD01
[SSDD VIL]	SSDD Virtual Interlocking	3BU_15000_6955_DSAPA	01 (planned)
[SSDD IPS]	System / Subsystem Design Description (SSDD) - IPS	3BU_17300_2052_DSAPA	01 (planned)
[SSDD ADM]	System / Subsystem Design Description (SSDD) – ADM	3BU_17300_2102_DSAPA	01 (planned)
[DoP DIL]	Docuplan - Generic Application - Digital Interlocking OBB	3BU_17300_1201_ADAPA	01 (planned)
[WPD EL BASE GENSYS&OBB]	WPD ELEKTRA Base Component PV_EL_GEN_2.20.0 & BL_OBB_40.00	https://e2.at.gts/confluence/x/i5NACQ 3BU_15000_6850_ADAPA	Latest 01
[WPD EL BASE CAR-RED]	WP ELEKTRA Base - Component CAR-RED	https://e2.at.gts/confluence/x/K8HVCg 3BU_15000_6800_ADAPA	Latest 01
[WPD VIL Testing]	WPD VIL Testing	https://e2.at.gts/confluence/x/le0bCw 3BU_15000_6957_ADAPA	Latest 02
[WPD Test Autom]	WP Description for Test Automation in DSTW context	https://e2.at.gts/confluence/x/BtJCCg 3BU_15000_6958_ADAPA	Latest 01
[WPD TAFE Convertor_1.0.0]	WPD TAFE Converter 1.0.0	https://e2.at.gts/confluence/x/tnOUCg 3BU_15000_6959_ADAPA	Latest 01
[WPD TAFE Convertor_1.1.0]	WPD TAFE Converter 1.1.0	https://e2.at.gts/confluence/x/uUleCw 3BU_15000_6962_ADAPA	Latest 01 planned
[WPD TAFE 1.2.1]	WPD TAFE 1.2.1	https://e2.at.gts/confluence/x/1C2UCg	Latest
[WPD TAFE 1.4.0]	WPD TAFE 1.4.0	https://e2.at.gts/confluence/x/Pk5eCw	Latest

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Id	Nr./Title	Reference	Version
[WPD OREST]	WPD OREST for Digital Interlocking	https://e2.at.gts/confluence/x/S54bCw 3BU_15000_6960_ADAPA	Latest 01
[WPD ECMUX]	WPD ECMUX	https://e2.at.gts/confluence/x/eqK5Bw 3BU_01500_6961_ADAPA	Latest 01
[TP]	Test Plan Solution Element Virtual Interlocking	3BU_15000_6955_DUAPA	01 (planned)
[TP DIL]	Test Plan Generic Application Digital Interlocking	3BU_17300_1216_QTAPC	01
[CMR Ed.03]	Change Management Report ELEKTRA Base GENSYS 2.20.0 & Test Automation	3BU_15000_6950_BCAPA	03
[SecTRA]	SECURITY THREAT & RISK ASSESSMENT (SECTA) - Generic Application Digital Interlocking ÖBB	3BU_17300_1217_DUAPA	01 (planned)
[Hitachi Glossary]	Hitachi Gloassary	3BU_15000_8008_WDAPQ	latest
[TeamCharterSEA]	Team Charter SEA	3BU_61400_0137_QMAPA	04
[TeamCharterSEV IL]	Team Charter Solution Element Virtual Interlocking	3BU_15000_6955_QMAPA	01 (planned)
[SysArch]	Top Level Architecture Drawing - Generic Application - Digital Interlocking OBB	3BU_17300_1218_DSAPA	02

2.2 PROCESS BASIS

The project is set up with the Management System, release number 2025-08 as process basis. Changes in this process basis (e.g. adaptation to a new release of Management System) have to be analysed, especially regarding the impact on already existing work products, and have to be documented and approved.

2.3 STANDARDS AND PROCEDURES

Table 9 : Standards and Procedures

Nr./Title	Publishing Date/Reference	Version/Revision
[S1] IRIS Certification rules:2023 based on ISO/TS 22163:2023	2023	n/a

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Nr./Title	Publishing Date/ Reference	Version/ Revision
[S2] EN 50716:2024 – Railway application – Requirements for software development	2024	n/a
[S3] EN 50129:2018 + AC:2019 Railway applications – Communication, signalling and processing systems – Safety related electronic systems for signalling	2018/2019	n/a
[S4] ISO 9001	2015	n/a
[S5] EN 50126-1:2017 – Railway Applications – The Specification and Demonstration of Reliability, Availability, Maintainability and Safety (RAMS) – Part 1: Generic RAMS Process	2017	n/a
[S6] EN 50126-2:2017 – Railway Applications – The Specification and Demonstration of Reliability, Availability, Maintainability and Safety (RAMS) – Part 2: Systems Approach to Safety	2017	n/a
[S7] Management System Process Manage Risks and Opportunities	Part of 87200984-00-PRJ-GRP-EN (Chapter 8.6)	2025-08
[S8] Management System Process Manage Project	87200984-00-PRJ-GRP-EN	2025-08
[S9] Minimum Requirements for the configuration management implementation	Part of 87100077-CFG-GRP	007
[S10] Management System Process 6.1 DDQS-SW - Design develop and qualify the Software solution	Part of 87202125-00-DDQ-GRP-EN	2025-08
[S11] Management System Process 6.1 DDQS-HW - Design develop and qualify the Hardware solution	Part of 87202126-00-DDQ-GRP-EN	2025-08
[S12] Management System Process Validate the Solution	Part of 87202127-00-DDQ-GRP-EN	2025-08
[S13] Management System Process Integrate and Verify the Solution	Part of 87202127-00-DDQ-GRP-EN	2025-08
[S14] Management System Process Design the Solution	Part of 87202127-00-DDQ-GRP-EN	2025-08
[S15] Management System Process 9.7 CFG - Manage Configuration	87202424-00-CFG-GRP-EN	2025-08
[S16] Management System Process 9.3 Control Export	87299999-00-EXP-GRP-EN	2025-08
[S17] Management System Guarantee quality and manage continuous improvement	87200214-00-QCI-GRP-EN	2025-08
[S18] Protection of Group Information	87201725-GOV-GRP-EN	005
[S19] Personal Data Protection Policy	87212828-GOV-GRP-EN	001
[S20] Project Life Cycle	83530275-DDQ-AUT-EN	005

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Nr./Title	Publishing Date/ Reference	Version/ Revision
[S21] Document Management Rules	83530061-DOC-AUT-EN	013
[S22] Milestone Template Solutions	83530019-DDQ-AUT-EN	012
[S23] Document List and Review Matrix for engineering projects	83530224-DOC-AUT-EN	011
[S24] HSE Instruction	83530048-HSE-AUT-DE	020
[S25] Agile Definition SMP	83530312-DDQ-AUT-EN	002
[S26] ISA/IEC 62443 - Security for industrial automation and control systems IEC 62443-3-3: System security requirements and security levels, Edition 1, 09 2013	2013	01
[S27] ISA/IEC 62443 - Security for industrial automation and control systems IEC 62443-4-1: Secure product development life-cycle requirements, Edition 1, 01 2018	2018	01
[S28] ISA/IEC 62443 - Security for industrial automation and control systems IEC 62443-4-2: Technical security requirements for IACS components, Edition 1, 04 2019	2019	01
[S29] EN 50159, + A1:2020 Railway applications – Communication, signalling and processing systems – Safety-related communication in closed transmission systems	2020	n/a
[S30] System Pillar Cybersecurity Specification	2025	001
[S31] CENELEC TS 50701:2023 Railway applications – Cybersecurity	2023	n/a
[S32] Template of DoX Templates	83530313-DDQ-AUT-EN	002

2.4 TERMS, ABBREVIATIONS AND ACRONYMS

2.4.1 ROLE ABBREVIATIONS

For role abbreviations see [Hitachi Glossary]. In addition, General Terms, Abbreviation and Acronyms that are not included in [Hitachi Glossary] see the following table:

Table 10 : Role abbreviations

ARC	Architect
CM	Configuration Manager
DM	Document Manager
HoD	Head of Discipline
IVM	Integration and Verification Manager

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IVVQM	Integration, Verification, Validation and Qualification Manager
PDA	Project Design Authority
PLA	Product Line Architect
PLM	Product Line Manager
PSM	Project Security Manager
QAM	Quality Assurance Manager
RAM-AM	Reliability, Availability, Maintainability Assurance Manager
RE	Requirements Engineer
RM	Requirements Manager
TM	Test Manager
SAM	Safety Assurance Manager
SecAM	Security Assurance Manager
SEM	Solution Engineering Manager
SM	Senior Management
VAL	Validator
VER	Verifier
WPM	Work Package Manager

2.4.2 GENERAL TERMS, ABBREVIATIONS AND ACRONYMS

For term and abbreviations see [Hitachi Glossary]. In addition, General Terms, Abbreviation and Acronyms that are not included in [Hitachi Glossary] yet see the following table:

Table 11 : General Terms, Abbreviation and Acronyms

BCWS	Budgeted Cost of Work Scheduled
CBB	Contract Budget Baseline
CCB	Configuration Control Board
CDR	Critical Design Review
CENELEC	European Committee for Electrotechnical Standardization
CHARM	Change and Request Management
CR	Change Request
DDQS	Design, Develop and Qualify the Solution
DWF	Digital Web Frontend of ClearCase Document Management
EDM	ELEKTRA Delivery Model

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EV	Earned Value
FEA-ID	Feature Identifier from Feature List
FQR	Final Qualification Review
GEN	Generic
HW	Hardware
IFS	Interfaces
IVV	Integration, Verification and Validation
KOM	Kick of Meeting
M	Mandatory
MPP	Manage Product Policy
N/A	Not applicable
O	Optional
PCM	Primavera Core Model
PCR	Problem Change Request
PDR	Preliminary Design Review
PEP	Personnel Efforts Planning
PiO	Put in Operation
PMP	Project Management Plan
QMP	Quality Management Plan
RAMS	Reliability, Availability, Maintainability and Safety
SE	Solution Element
SEMP	System Engineering Management Plan
SFR	Solution Functional Review
SIL	Safety Integrity Level
SOR	Solution Orientation Review
SRR	Solution Requirement Review
SW	Software
TQR	Technical Qualification Review
TRR	Test Readiness Review
WBS	Work Breakdown Structure
WP	Work Package
WPD	Work Package Description

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2.4.3 PROJECT SPECIFIC TERMS, ABBREVIATIONS AND ACRONYMS

For Specific Terms, Abbreviation and Acronyms see [Hitachi Glossary]. In addition, Specific Terms, Abbreviation and Acronyms that are not included in [Hitachi Glossary] yet see the following table:

Table 12 : Specific Terms, Abbreviation and Acronyms

ADM	Administration Diagnostic Monitoring
AzLM	Multi Section Axel Counter (one type of TDS)
BI	Basic Integrity
B-Vi	Basic Virtualization
CFI	Customer Furnished Item
DC	Data Centre
DCA	Distributed Controller Architecture
DDSP	Daten Diagnose Service Plattform (CFI ÖBB)
DFE	Digital Field Elements
DGP	Diagnosis Processor (ELEKTRA)
DIL	Digital Interlocking = DSTW
DSTW	Digitales Stellwerk = DIL
DSTW-ZE	Digitales Stellwerk Zentraleinheit = EIL
ECMUX	Element Controller MultipleXer
EIL	Electronic Interlocking System according to EULYNX – equals to DSTW-ZE
EKSA	Eisenbahnkreuzungssicherungsanlage = LCSi (CFI ÖBB)
EULYNX	European Initiative to Linking Interlocking Systems
FES	Field Element Station
GA DIL OBB	Generic Application Digital Interlocking ÖBB
IEC	International Electrotechnical Commission
IPS	Integrated Platform Services (Integrierte Plattform Sicherheitstechnik)
IPSS	IPS Solution Element
IXL	Interlocking
LCSi	Level Crossing Safety Installation = EKSA
LX	Level Crossing
MDM	EULYNX, Maintenance and Data Management

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ÖBB	Österreichische Bundes Bahn (Austrian Railway Duty Holder, ÖBB)
OC	Object Controller
OMC	Operation Management Center = BFZ (Betriebsführungszentrale)
PZB	Punktförmige Zugbeeinflussung = Intermittent Train Control System
RaSTA	Rail Safe Transport Application (safety protocol for railway signalling systems)
RBC	Radio Block Center (CFI ÖBB)
RCU	Remote Controller Unit
SCI	Standard Communication Interface (EULYNX interface for process data information)
SCWS	Signal Controlled Warning System
SDI	Standard Diagnostic Interface (EULYNX interface for diagnostics)
SiDiNet	SicherungsDienstNetzwerk (CFI ÖBB)
SMI	Standard Maintenance Interface (EULYNX interface for maintenance)
Solution	A consistent set of systems, equipment and services provided to the Customer to meet a specific requirement
Solution Element	A system, a sub-system, a product / equipment, or a service, a software component, a hardware component
SSI	Standard Security Interface (EULYNX interface for security)
SSP	Security Services Platform (CFI ÖBB)
TDS	Train detection system
TMS	Traffic Management System
VIL	Virtual Interlocking
VSTW	Virtuelles Stellwerk = VIL

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3. SOLUTION ELEMENT ENGINEERING AND DEVELOPMENT PROCESS

The Solution will be conducted according to Management System 2025-08. As the Management System is continuously improving, it has to be mentioned, that in the evolution of the Solution, for newly created documents within this project, the respective valid templates have to be used according to the up-to-date Management System 2025-08 version. For maintenance and extension of existing documents the previously used template may remain.

3.1 ORGANIZATION OF SOLUTION ELEMENT ENGINEERING

The description of the roles mentioned below can be found on the [Management System] Portal & agile roles mentioned in [Agile Definition SMP].

3.1.1 SOLUTION ELEMENT CORE TEAM

Table 13: Solution Element Core team according to Management System

Role	Resource	Deputy Resource
GEN SEM	E. Stemmer	M. Gartner-Pohl
GEN PDA	M. Eichberger	
GEN IVM	NN	
PLM	R. Holz knecht	TBD
SEM	P. Bargiel	B. König
PDA ELEKTRA	E. Glaser	TBD
PDA ECMUX	R. Holzmann	TBD
PDA CAE DATA Tools	S. Krasny	James Saunders
IVM / WPM VIL Testing	D. Olah	TBD
VER	H. Walenta	E. Rahmani
SAM	R. Büger	K. Radda
PSM	T. Fischer	TBD
QAM	C. Janneschitz	F. Rotter
VAL	I. Grgic	P. Satlek
RM	J. Mach	G. Eber
WPM ELEKTRA BASE	B. König	P. Bargiel
WPM ELEKTRA OBB	B. König	P. Bargiel
WPM ECMUX	R. Rambo	P. Bargiel
WPM CAE Data Tools	R. Ungerboeck	S. Krasny

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WPM ELEKTRA Test Automation	A. Chita	P. Bargiel
WPM Test Automation Framework & Convertor	E. Diethelm	M. Weigensam
PLA	A. Paul	-
ARC ELEKTRA	T. Reitmayr W. Arthold	-
ARC ECMUX	R. Holzmann	K. Aichner
ARC CAE Data Tools	J. Saunders	TBD
CM	M. Socha	TBD
SecAM	A. Willenberg	TBD
DM	B. König, see *) comment below	P. Bargiel
RAM-AM	F. Moser	TBD
Integration	T. Pallierer	T. Reitmayr

*) The role DM is not executed by a single person. For details please refer to chapter [5.5.4] Role of DM

**) The verification tasks will be handled according to CENELEC [S2] EN 50716:2024 – Railway application – Requirements for software development. The review of all documents is performed through reviews by the roles defined in the Document List and Review Matrix for engineering projects [S23]. For details please refer to chapter 5.5.3 Role of Verifier (VER) according to CENELEC EN 50716 and EN 50129

3.1.2 SOLUTION ELEMENT CORE AGILE TEAM

Table 14: Agile roles according to [Agile Definition SMP].

Role	Resource	Deputy Resource
PO Solution Element VIL	P. Bargiel	B. König
Technical Lead Solution Element VIL	E. Glaser	P. Bargiel
PO VIL Testing	D. Olah	TBD
Technical Lead Testing	D. Olah	TBD
PO ELEKTRA BASE	B. König	P. Bargiel
Technical Lead ELEKTRA BASE	E. Glaser	T. Reitmayr, W. Arthold
PO ELEKTRA OBB	B. König	P. Bargiel
Technical Lead OBB	E. Glaser	T. Reitmayr, W. Arthold
PO ELEKTRA BASE CAR-RED	P. Bargiel	E. Glaser
Technical Lead CAR-RED	E. Glaser	M. Paslariu

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PO ECMUX	R. Rambo	P. Bargiel
Technical Lead ECMUX	R. Holzmann	K. Aichner
PO ELEKTRA Test Automation	A. Chita	D. Olah
Technical Lead Test Automation	A. Chita	D. Olah
PO Test Automation Framework	E. Diethelm	M. Weigensam
Technical Lead Automation Framework	E. Diethelm	M. Weigensam

3.1.3 WORK PACKAGES TEAM

The description of the work packages team can be found in the corresponding work package description: [WPD EL BASE CAR-RED], [WPD EL BASE GENSYS&OBB], [WPD TAFE 1.2.1], [WPD TAFE Convertor_1.0.0], [WPD TAFE 1.2.1], [WPD Test Autom] & [WPD VIL Testing].

3.1.4 EXTERNAL PROJECT ROLES

External project roles are managed by Generic Application Digital Interlocking OBB and described in [SEMP DIL] & [PMP].

3.1.5 CONFIGURATION CONTROL BOARD (CCB)

Table 15: CCB

Role	Resource	Comment	M/O
PLM	see Table 13		M
PM	H. Wolfram	Budget, schedule, scope	M
SEM	see Table 13	Budget, schedule, scope	M
GEN SEM	see Table 13	Budget, schedule, scope	M
PDA ELEKTRA	see Table 13		M
PDA ECMUX	see Table 13	Scope ECMUX	O
PDA CAE DATA Tools	see Table 13	Scope CAE DATA Tools	O
GEN PDA	see Table 13		M
SAM	see Table 13		M
IVM	see Table 13		M
GEN IVM	see Table 13		M
VAL	see Table 13		O

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QAM	see Table 13		M
PSM	see Table 13		O
WPM ELEKTRA BASE	see Table 13	Budget, schedule ELEKTRA BASE	M
WPM ELEKTRA OBB	see Table 13	Budget, schedule ELEKTRA OBB	M
WPM ECMUX	see Table 13	Budget, schedule ECMUX	M
WPM CAE Data Tools	see Table 13	Budget, schedule CAE Data Tools	M
WPM ELEKTRA System Test Automation	see Table 13	Budget, schedule Test Automation	O
WPM Test Automation Framework	see Table 13	Budget, schedule Test Automation Framework	O
Head of Engineering	M. Vana	for priority conflicts regarding resources in E & Budget of Project ID 100020889	O

Other project team members are invited to the CCB as needed. M = mandatory, O = optional

3.1.6 SUBCONTRACTORS

- Sub-contractions for Pen test (and organization, performing) will be handled at Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0) for internal and external Pen test, see for details [SEMP DIL]. Preconditions are stated in [WPD VIL Testing].
- Sub-contractions with Hitachi Spain for DCA 3.1.0 will be handled at Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0). Preconditions are stated in [WPD ECMUX].
- Sub-contractions with Hitachi Germany for AZLM 7.10 will be handled at Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0). Preconditions are stated in [WPD ECMUX] & [WPD OREST].

3.2 SYSTEM ENGINEERING SUPPLIES

Table 16: System Engineering Suppliers

Item	Version	Provider
IPS	IPS 40.0.0	Hitachi Rail Austria
ADM	ADM 40.0.0	Hitachi Rail Austria
MiniOMC	MiniOMC 40.0.0	Hitachi Rail Austria
DCA	DCA 3.1.0	Hitachi Rail Spain
AZLM	AZLM 7.10	Hitachi Rail Germany
TAS Platform Core	2.6.1	Hitachi Rail Austria
TAS Platform OCS	3.6.1	Hitachi Rail Austria
TAS Platform MNT	3.6.1	Hitachi Rail Austria
TAS Platform J4S	3.6.0	Hitachi Rail Austria

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RJC RASTA	2.1.0	Hitachi Rail Austria
RJC Core	1.2.0	Hitachi Rail Austria
ELEKTRA OBB	OBB_29.40	Hitachi Rail Austria
GNU Chill	2.3.13	Hitachi Rail Austria

3.3 INPUT DATA

3.3.1 SOLUTION ELEMENT INPUT DATA

See also Documentation Plan [DoP].

The SEMP describes the application of the Management System (Release from 2025-08) to the design, development and integration of the Solution Element Virtual Interlocking OBB Phase 1 (VIL OBB 1.0.0). The following table shows the most relevant input data:

Table 17: Input Data

Title	Description	Ref.	Provider
<i>Solution Management Plan</i>	Focuses on the Solution Management of Solution Generic Application System including organization, policies and plans that are carried out by Hitachi Rail AT in relation to the Project	[SEMP DIL]	Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0)
<i>Solution Element Management Plan IPSS</i>	Focuses on the Solution Management of Solution Element IPSS including organization, policies and plans that are carried out by Hitachi Rail AT in relation to the Project	[SEMP SE IPSS]	Solution Element IPS Solution Phase 1 (IPSS OBB 1.0.0)
<i>Solution Element Management Plan DFE</i>	Focuses on the Solution Management of Solution Element DFE including organization, policies and plans that are carried out by Hitachi Rail AT in relation to the Project	[SEMP SE DFE]	Solution Element Digital Field Elements Phase 1 (DFE 1.0.0)
<i>Project Management Plan</i>	Focuses on the Management System including organization, policies and plans that are carried out by Hitachi Rail AT in relation to the Project.	[PMP]	Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0)
<i>Configuration Management Plan</i>	Provides details of the formal Configuration Management process.	[CMP]	Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0)
<i>Quality Management Plan DIL</i>	Describes requirements for all quality assurance measures and quality assurance activities to be carried out in	[QMP DIL]	Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0)

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Title	Description	Ref.	Provider
	the solution as contribution to the fulfilment of the project goals.		

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Title	Description	Ref.	Provider
<i>RAM Plan</i>	Provides details of the tasks and activities about Reliability, Availability and Maintainability of the Solution. Project specific RAM Plans per solution element.	[RAMP GEN]	Generic Application Digital Interlocking ÖBB 1.0.0 (GA DIL ÖBB 1.0.0)
<i>Safety Plan</i>	Provides details of the tasks and activities of Solution safety management and Solution safety engineering required to identify, evaluate and eliminate or control hazards throughout the life cycle. Reference to Project Specific Safety Plan	[PSP]	Generic Application Digital Interlocking ÖBB 1.0.0 (GA DIL ÖBB 1.0.0)
<i>Test Plan DIL</i>	Describes Collaborative Labor Environment Setup and maintenance activities for tests according to ONE lab approach for Solution Element. Supporting and coordination activities. External penetration tests performed by an external company	[TP DIL]	Generic Application Digital Interlocking ÖBB 1.0.0 (GA DIL ÖBB 1.0.0)
<i>IV Plan</i>	Is the general overview how it is planned to manage and test the Generic Application with the three Solution Elements. The project specific plan contains the ÖBB requirements and is named V&V plan	[IVP]	Generic Application Digital Interlocking ÖBB 1.0.0 (GA DIL ÖBB 1.0.0)
<i>VQ Plan</i>	Is the general overview how it is planned to manage and validate the Generic Application with the three Solution Elements. The generic VQP and a project specific plan following the ÖBB requirements is provided	[VQP]	Generic Application Digital Interlocking ÖBB 1.0.0 (GA DIL ÖBB 1.0.0)
<i>Security Management Plan</i>	Is the general overview how it is planned to address the security aspects in the Generic Application with the three Solution Elements. The generic SecMP and a project specific plan following the ÖBB requirements is provided	[SecMP]	Generic Application Digital Interlocking ÖBB 1.0.0 (GA DIL ÖBB 1.0.0)
<i>Requirement Management Plan</i>	Describes the requirement artefacts, requirement types, requirement attributes, the tracing, the requirements management process and the metrics and tools to be used for measuring, reporting, and controlling changes to the requirements. It further describes how the project will set up requirement documents, requirement types, their respective requirement attributes and traceability.	[RMP]	Generic Application Digital Interlocking ÖBB 1.0.0 (GA DIL ÖBB 1.0.0)

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Title	Description	Ref.	Provider
<i>Customer Requirements Specification</i>	Set of documents Pflichtenheft provided by the Customer OBB	[CRS]	OBB
<i>System Segment Specification</i>	Is the general overview about the requirements to address the system aspects in the Generic Application contains allocation to the Subsystem Solution Element Virtual Interlocking OBB Phase 1 (VIL OBB 1.0.0) and to components CAR-RED_1.0.0, EEC SW OBB 04.00, ELEKTRA BL OBB 40.00, ELEKTRA PV GEN 2.20.0, OREST 19.50	[SSS DIL]	Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0)
<i>System Segment Security Specification</i>	Is the general overview about the requirements to address the security aspects in the Generic Application contains allocation to the Subsystem Solution Element Virtual Interlocking OBB Phase 1 (VIL OBB 1.0.0) and to components CAR-RED_1.0.0, EEC SW OBB 04.00, ELEKTRA BL OBB 40.00, ELEKTRA PV GEN 2.20.0, OREST 19.50	[SSS SEC DIL]	Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0)
<i>System Segment Design Description</i>	Contains the top level architecture description Generic Application and contains design description to the Subsystem Solution Element Virtual Interlocking OBB Phase 1 (VIL OBB 1.0.0) and to components CAR-RED_1.0.0, EEC SW OBB 04.00, ELEKTRA BL OBB 40.00, ELEKTRA PV GEN 2.20.0, OREST 19.50	[SSDD DIL]	Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0)
<i>Feature List DIL</i>	Is the general overview about the features to address the system and subsystem aspects in the Generic Application contains allocation to the Subsystem Solution Element Virtual Interlocking OBB Phase 1 (VIL OBB 1.0.0) and to components CAR-RED_1.0.0, EEC SW OBB 04.00, ELEKTRA BL OBB 40.00, ELEKTRA PV GEN 2.20.0, OREST 19.50	[MCRD DIL]	Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0)
<i>Feature List OBB</i>	Is the general overview about the features to address in the component ELEKTRA BL OBB 40.00	[MCRD OBB]	ELEKTRA BL OBB 29.50

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Title	Description	Ref.	Provider
<i>Team Charter</i>	Team Charter SEA describes the way how the EPR SEA team organises itself to handle all SEA. SW related activities in an agile way tailoring the Agile Definition SMP [S25]. It is not dedicated to a specific project but focuses on the working model for the SEA team.	[TeamCharterSEA]	HoEPR
<i>Team Charter</i>	Team Charter GenSys Engineering describes the way how the EPR GenSys Engineering team organises itself to handle all SW related activities in an agile way implementing the Agile Definition SMP [S25]. It is dedicated to Solution Element Virtual Interlocking OBB Phase 1 (VIL OBB 1.0.0) project and focuses on the working model for the GenSys Engineering team.	[TeamCharterSEVIL].	SEM VIL

3.3.2 WORK PACKAGE INPUT DATA

Additional Input data of the component level handled as work packages of this Solution Element please refer to Work package descriptions in mentioned in chapter [1.1.4].

3.4 PRECONDITIONS TO OTHER RELATED SOLUTIONS OR SOLUTION ELEMENTS

3.4.1 PRECONDITIONS OF SOLUTION ELEMENT

General precondition for the Solution Element Virtual Interlocking OBB Phase 1 (VIL OBB 1.0.0):

Table 18: Preconditions

ID	Deliverable	Target Date	Provider / comment
VIL_P01	Contract Signature with OBB	12/2025	OBB For latest available dates see [IMP].
VIL_P02	Release of KOM DIL document set 1) [MCRD OBB] 2) [MCRD OBB]	TBD	Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0) For latest available dates see [IMP].

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VIL_P03	Release of DIL document set for VIL SOR2 1) [SEMP DIL] 2) [PMP] 3) [CMP] 4) [PSP] 5) [TP DIL] 6) [IVP] 7) [VQP] 8) [SecMP] 9) [RMP] 10) [IMP] 11) Operational Concept Description (OCD) 12) Product Breakdown Structure (PBS) 13) Cost Breakdown Structure (CBS) 14) Security Risk Assessment Plan (SecRAP) 15) Security Risk Assessment Report (SecRAR) 16) Integrated Master Schedule (IMP)	Right after KOM DIL / 01/2026	Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0) For latest available dates see [IMP]. not Referenced documents in section [2.1] see [DoP DIL]
MS E03	Pflichtenheft Freigabe from Customer OBB defined in	04/2026	OBB Precondition to perform Millstone SRR2 Defined in [PMP]. For latest available dates see [IMP].
VIL_P05	Release of document set [SSS DIL] & [SSS SEC DIL].	after MS E03 + 2 months	Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0)
VIL_P06	Release of [SSDD DIL]	after MS E03 + 4 months	Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0) Precondition to perform SFR
VIL_P07	Release of DIL launch review document set 1) Business Case (BC) 2) Product Plan	06/2026	Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0) / Precondition to perform SFR

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VIL_P08	TRR1 IPSS OBB 1.0.0	Xx/202x	Solution Element IPSS OBB 1.0.0 / To start final validation in WP VIL Testing
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3.4.2 PRECONDITIONS OF WORK PACKAGES

Preconditions of the component level handled as work packages of this Solution Element please refer to Work package descriptions in mentioned in chapter [1.1.4].

3.5 TECHNICAL OBJECTIVES

The requirements basis is described in [RMP] and document sets for the Solution Element Virtual Interlocking OBB Phase 1 (VIL OBB 1.0.0) are stated in section Input data [3.3] the delivery dates for the input data is stated in section Preconditions of solution Element [3.4.1]. All requirements allocated to Solution Element Virtual Interlocking OBB Phase 1 (VIL OBB 1.0.0) or Work Packages mentioned in section [1.1.4] are representing the technical objectives for this solution element.

3.6 DETAILED FIRST LEVEL WBS FOR WORK PACKAGES

Detailed first level WBS for the work packed can be found in [WBS&OBS]

3.7 SPECIAL NEED FOR TRAINING OF SYSTEM ENGINEERING TEAM

The Solution Element Virtual Interlocking OBB Phase 1 (VIL OBB 1.0.0) will contribute to the training plan in [SEMP DIL]. In addition, specific training areas in Solution Element Virtual Interlocking OBB Phase 1 (VIL OBB 1.0.0) will be

- Agile Basics
- CENELEC EN 50716:2023
- Railway Domain Knowledge | Austria
- TAS Plf. NMR
- TAFE
- GIT
- Virtual Elektra
- Eclipse
- maven
- Security
- PCM
- Polarion

3.8 CONSTRAINTS / RISKS

Constraints are already considered in the [RiskReport].

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3.9 WORK CONTAINED

- 1) The subsystem managed as Solution Element Virtual Interlocking OBB Phase 1 (VIL OBB 1.0.0) with focus on the Phase 1 minimum technical objective “Basic Virtualization” (B-Vi) see section [3.5] by this document with the
 - a) purpose and scope described in section “Solution Element of Solution” [1.1.2]
 - b) Milestones for Solution Element described in section [1.3.1] and
 - c) Delivery of the items listed in section Solution Element Deliverables [1.3.2].
 - d) Support to Input data mentioned in section 3.3
- 2) The specification, development of components of this subsystem managed as software components:
 - a) Work Package ELEKTRA Base Phase 0 & Phase 1 - Component GENSYS_2.20.0 described in [WPD EL BASE GENSYS&OBB] covers the development of the generic software part of the ELEKTRA Core illustrated in Figure 3: Digital Interlocking System Architecture. This work package also includes the integration of subcomponent CAR-RED_1.0.0.
 - b) Work Package ELEKTRA Base Phase 0 & Phase 1 - Component CAR-RED_1.0.0 described in [WPD EL BASE CAR-RED] covers the development of the elements CAR library, the EP handler library, C headers and CHILL grant files, the CAR console, configuration templates, and various tools. These components support the CHILL Application Runtime (CAR) Redundant (RED), which runs on a redundant TAS Platform Core setup in hot-hot mode and is situated in software part of the ELEKTRA Core illustrated in Figure 3: Digital Interlocking System Architecture.
 - c) Work Package ELEKTRA ÖBB Phase 1 covers the development of the country specific software part of the ELEKTRA Core illustrated in Figure 3: Digital Interlocking System Architecture. This work package also includes the integration of Component ELEKTRA PV GEN 2.20.0.
 - d) The work package Work Package ECMUX Phase 1 described in [WPD ECMUX] covers SW development for component ECMUX illustrated in Figure 3: Digital Interlocking System Architecture and contains the development of elements IO Generic (DispatchLamp, UnivElem), Level Crossing (EULYNX), ÖBB Light Signals, Keylocks, Power Supply, Techbase (ELEKTRA CC 2oo2), and Security, TDS (EULYNX)/AzLM, IO Generic (simulated), and Point integration with OC-P MAV2/3.
- 3) An CAE Data Preparation Tool is enhanced in Work Package CAE Data Tools Phase 1 for the Subsystem ELEKTRA PV GEN 2.20.0, CAR-RED_1.0.0, ELEKTRABL OBB 40.00 EEC SW OBB 04.00
- 4) The integration and verification of this subsystem managed by work packages
 - a) Work Package VIL Testing Phase 1 described in [WPD VIL Testing] with scope of planning, progress and documentation of the integration and system testing activities including CCB Sub IT testing, VIL Integration tests including, VIL SysAdmin Tests, VIL Internal Interaction tests, VIL EULYNX OC Integration tests - in a cooperation with ECMUX WP, VIL X25overRaSTA Integration tests (interaction with Adjacent Systems), VIL Basic ADM Integration tests, VIL System testing - manual execution, VIL Performance Tests, VIL Security Testing, corresponding Labor and Test Environment Setup for VIL EULYNX OC Integration tests and maintenance activities - in the cooperation with DIL according to the ONE LAB approach & supporting activities for DIL integration and the IPS Solution/ADM integration. This work package also includes the integration of work package Work Package ELEKTRA ÖBB Phase 1 & Work Package ECMUX Phase 1
 - b) Supported by subsystem test automation Work Package ELEKTRA Test Automation Phase 0 & Phase 1 increasing the percent of the automated test case to 80% of TCS System Test applicable to Work Package VIL Testing Phase 1 with the support of enabling tools Work Package Test Automation Framework Convertor & Work Package Test Automation Framework Phase 1

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3.10 WORK NOT CONTAINED

While the work contained is already stated in section [3.9], the following topics are explicitly mentioned as not part of this solution in order to sharpen the responsibility:

- 1) Any software, product or solution development of supplying this solution element described in section System engineering supplies [3.2]
- 2) Any software, product or solution development mentioned in [SEMP SE DFE] and [SEMP SE IPSS].
- 3) Any software, product or solution development integrate, test, verify, validate and assess of Phase 2 and Phase 3 activities not mentioned in [SEMP DIL].
- 4) Assess the solution element see [1.1.2] is part of [SEMP DIL].
- 5) Industrialization of HW components (see [SEMP SE DFE])
- 6) Execution of any field tests on site
- 7) Execution of customer acceptance tests (details see [IVP] and [SEMP DIL] WP description WP4)
- 8) Coordination and execution of Factory Acceptance Tests (Customer lab-/integration-tests; (details see [IVP] and [SEMP DIL] WP description WP4))
- 9) Installation and put into operation at customer site
- 10) Additional features as offered to ÖBB like Geo-redundancy, E-Shu, ...
- 11) The integration of third-party OCs – however the integration activity is a clear request of the customer and can be requested by a separated purchase order
- 12) Backward compatibility/supporting previous generation Elektra projects, however reused artifacts would be handled in a way that beneficial for previous projects as well
- 13) VIL solution is also not responsible to integrate IPS/ADM with VIL (testing all scenarios) notice: VIL is using integrating with IPS on a single VIL level.

3.11 RESPONSIBILITY ASSIGNMENT MATRIX

The responsibilities of the roles at a Milestone Review can be seen in [MS-Report].

This Solution Element Virtual Interlocking OBB Phase 1 (VIL OBB 1.0.0) support the milestone review TQR2 together with Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0) which confirms the successful completion of product validation and the readiness of all needed elements for customer acceptance and that the Solution is ready for Assessment according to CENELEC. The PDAs confirms with the approval to milestone TQR2 that the delivered solution meets his/her expectations.

This Solution Element Virtual Interlocking OBB Phase 1 (VIL OBB 1.0.0) support the milestone review FQR together with Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0) confirms that the Design, Development and Qualification activities including the support to assessment according to CENELEC so Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0) is ready for deployment.

3.12 TOOLS AND FACILITIES

The following subsections provide a rough overview of the project tool setup. For version information and tool classification see document [ToolClassSW].

In this chapter the needed tools for executing the Solution Element Virtual Interlocking OBB Phase 1 (VIL OBB 1.0.0) will be identified.

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This Table give information about planning information (need for a license, cost of the license, need for a special hardware, and need for user training, alternatives ...) which tools are used/not used/not applicable:

Table 19: Tools and Facilities

Tool Category	Tool	Note User max/concurrent
Project Management	1) Primavera PCM 2) Advanced Roadmaps for Jira, Jira Scrum Report functions	1. WPMs+ SEM: 6 users 2. WPMs + SEM: 6 users
Task Management	Jira iTask	All: 80 users
Requirements Management	DOORS/RMF, Polarion	SEM, IVM, PSM, RM, PDA, ARC, WPMs, VAL: 26 users
Management Traceability	DOORS/RMF, Polarion	SEM, IVM, PSM, RM, PDA, SwE, Tester, ARC, WPMs, Designer: 60 users
Configuration Management	1) ClearCase 2) Git/Bitbucket	1) All WP with ClearCase: 40 users 2) All WPs with Git/Bitbucket: 40 users
Change Management	CHARM	All: 80 users
Documentation Management	1) ClearCase 2) DWF 3) SharePoint 4) Confluence	1) All: 80 users 2) All: 80 users 3) All: 80 users 4) All: 80 users
Documentation Generation	1) DOORS/RMF 2) Polarion 3) MS Office	1) All: 80 users 2) All: 80 users 3) All: 80 users
IVV Management	1) Polarion 2) Jira iTask 3) Confluence 4) MS Excel	1) SEM, IVM, VER, Tester: 23 User 2) SEM, IVM, VER, Tester: 23 User 3) SEM, IVM, VER, Tester: 23 User 4) SEM, IVM, VER, Tester: 23 User

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Code Building Tools	1) Jenkins 2) Maven 3) Nexus 4) ClearMake 5) ReviewBoard 6) SonarQube	1) WP ECMUX Team: 18 User 2) WP ECMUX Team: 18 User 3) WP ECMUX Team: 18 User 4) TBA 5) 8 User 6) User?
Coverity for code checks	?	?
Model Driven Engineering	Capella	User how many?
Hitachi AT Internal Used Products	? e.g. Data Prep. Tool AWS Data Prep. Verif. Tool AWS ELEPRO (Seibersdorf) Orest TAS Platform GNUCHILL	
Development Platform	Eclipse ? e.g. Linux/RedHat Linux/Ubuntu Windows	
Knowledge Base, Decision and Meeting Records	Confluence iTASK	
Tool category Document Review tool	Please review	
Security Scanners	TBA	

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4. PROCESS CONTROL

4.1 CONTROL OF SOLUTION DEVELOPMENT

The components and work packages and assigned responsibilities are listed in Table 4.

4.1.1 DEVIATION

The development of a Solution has to be planned and executed according to the DDQS Processes of Management System & Agile Definition [S25]. Deviation to the description in Project Life Cycle [S20] & Agile Definition [S25] is described in section [5].

4.1.2 METRICS

This Solution Element Virtual Interlocking OBB Phase 1 (VIL OBB 1.0.0) consists of two project ID's. The used metrics to control the engineering process are identified in the two different Engineering Dashboards. Several Components / Work packages listed in Table 4 and management activities listed in Table 3 are managed under a separate Project ID. The mapping can be found in the following Table 5: WBS mapping to Project IDs. As a result the metrics can be found under the following Engineering Dashboards:

Table 20: Mapping of Engineering Dashboard to Project ID

Project ID	Engineering Dashboard reference
100025795 / ATRELDE91	[EngDB SE VIL]
100020889 / ATRENDB35	[EngDB EL BASE & TA]

The tailoring for the dashboards is listed below:

Table 21: Metrics for [EngDB SE VIL]

Sheet	Tailoring
☒ Overview	Used - no deviation.
☒ Project Status	Used - no deviation.
☒ [01] Master Schedule	Master schedule for entire Solution element based on [IMP]. Separate Separate schedules for components / work packages based on work package description of section [1.1.4]
☒ [02] Milestones	Used - no deviation.
☐ [03] Workload	Not used. Reporting over [EngDB DIL].
☒ [04] Technical Risks	Used - no deviation
☒ [06] Project Burnup	Using of JIRA iTask Velocity Charts and JIRA iTask Version Report instead of Agile Metrics Template

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☐	05 Finance	Not used. Reporting over [EngDB DIL]. See Note 1.
☒	07 Quality Advice	Used - no deviation
☒	08 IVV Testing	Test case writing status is only reported on the corresponding components / work packages to this dashboard, see Table 5: WBS mapping to Project IDs & Table 20: Mapping of Engineering Dashboard to Project ID
☒	10 OBS	Used - no deviation
☒	11 WBS	Used - no deviation
☒	12 Documentation	Used - no deviation
☒	15 Security	Used - no deviation

Table 22: Metrics for [EngDB EL BASE & TA]

Sheet	Tailoring
☒ Overview	Used - no deviation.
☒ Project Status	Used - no deviation.
☒ 01 Master Schedule	Master schedule for entire Solution element based on [IMP]. Separate Separate schedules for components / work packages based on work package description of section [1.1.4]
☒ 02 Milestones	Used - no deviation.
☒ 03 Workload	Used - no deviation.
☒ 04 Technical Risks	Used - no deviation
☒ 06 Project Burnup	Using of JIRA iTask Velocity Charts and JIRA iTask Version Report instead of Agile Metrics Template
☒ 05 Finance	Used - no deviation. See Note 1.
☒ 07 Quality Advice	Used - no deviation
☒ 08 IVV Testing	Test case execution status is only reported on the corresponding components / work packages to this dashboard, see Table 5: WBS mapping to Project IDs & Table 20: Mapping of Engineering Dashboard to Project ID
☒ 10 OBS	Used - no deviation
☒ 11 WBS	Used - no deviation
☐ 12 Documentation	Not used. Report over [EngDB SE VIL]
☐ 15 Security	Not used. Report over [EngDB SE VIL]

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Note 1: ⓘ Guideline:

- a) “BCWS Planned Spend” is the monthly planned costs of work. This value can be determined from PCM. It is the defined/updated at the time the CBB is defined or updated.
- b) “Forecast” is the spending forecast from the last month with actuals to the end of the project (before this period it has to be empty).
- c) “EV” is not used.

Note 2: ⓘ Guideline:

The monitoring of project progress within an agile development environment requires a set of metrics defined in the Agile Definition SMP [S25] document sections 9.2.1 ‘Stability of Team Velocity’ and 9.2.2 ‘Visibility Point Burnup’ and is expected by senior management to be displayed during the PPPRs.

Furthermore, the Tailoring of the monitoring of project progress within an agile development environment in section [5.6] must be taken into consideration to get an adequate information basis for the KPIs.

4.2 MONITORING AND CONTROL OF SOLUTION ENGINEERING ACTIVITIES

4.2.1 PERIODIC INTERNAL REVIEWS AND MEETINGS

- Solution Element Virtual Interlocking Phase 1 Management
- Work Package ECMUX Phase 1
- Work Package CAE Data Tools Phase 1
- Work Package VIL Testing Phase 1
- ELEKTRA BASE Phase 0 & Phase 1 Management
- Work Package ELEKTRA Base Phase 0 & Phase 1 - Component GENSYS_2.20.0
- Work Package ELEKTRA Base Phase 0 & Phase 1 - Component CAR-RED_1.0.0
- Work Package ELEKTRA ÖBB Phase 1
- Work Package ELEKTRA Test Automation Phase 0 & Phase 1
- Work Package Test Automation Framework Phase 1
- Work Package Test Automation Framework Convertor

Table 23: Reviews and Meetings

Meeting	Responsible	Attendees	Frequency	Objectives
SE VIL Product Increment Synchronization Meeting	SEM	WPMs/POs, SEM	Every 3 weeks	• Alignment on solution element level on timeline, deliverables, dependencies for upcoming sprints • Align & Define goals in next sprint

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Meeting	Responsible	Attendees	Frequency	Objectives
Sprint start / kick off	WPMs/POs	Attendees following the Agile Definition [S25].	Every 3 weeks	Objectives following Agile Definition [S25]. The work packages have a separate timeboxed appointment for sprint start, review or retrospective in a very prompt timeframe of 1-2 days. Applied on: - Solution Element Virtual Interlocking Phase 1 Management - Work Package ECMUX Phase 1 - Work Package VIL Testing Phase 1 - Work Package ELEKTRA Base Phase 0 & Phase 1 - Component GENSYS_2.20.0 - Work Package ELEKTRA Base Phase 0 & Phase 1 - Component CAR-RED_1.0.0 - Work Package ELEKTRA ÖBB Phase 1 - Work Package ELEKTRA Test Automation Phase 0 & Phase 1
Sprint Review per component / Work package	WPMs/POs		Every 3 weeks	
Timeboxed Backlog Refinement per EPIC	WPMs/POs		On demand	
Timeboxed Retrospective	ScrumMaster		Every 3 weeks	
Periodically review of EngDB input by WPMs	SEM	WPMs	monthly	Review on report to EngDB progress, KPI dashboard, Progress Matrix, required decisions from WPMs.
Periodically Performance Project Solution Element Review (PPPR)	SEM	DIL SEM, QAM, DIL PM	monthly	Report of costs and progress, KPI dashboard, Progress Matrix, required decisions from SM to HoD Project Engineering, HoD Engineering
WPM alignment	SEM	WPMs	On demand	Alignment on costs and progress, KPI dashboard, Progress Matrix, required decisions from WPMs
SEM alignment	DIL SEM	SEM, IVM and SEMs of Solution Elements	On demand	Alignment on solution level on timeline, deliverables, dependencies

4.2.2 MILESTONE REVIEWS AND AUDITS

Milestone reviews are done according to Milestone Template Solutions [S22], result can be found in [MS-Report]. Quality Audits will be conducted through the QAM – see [QMP DIL]. Safety Audits will be conducted through the SAM – see [PSP].

Preliminary Design Review (PDR) & Critical Design Review (CDR) are not used and tailored see section [5.4] along the agile development method applied in Engineering for small and medium sized projects as

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described in Agile Definition SMP [S25]. Additional deliverables Visibility Points in design and implementation with deliverables monitored in chapter [1.3.2] & interactive check on EPICs within the Work packages following the Agile Definition [S25] & Definition of EPIC Done Template following [S32].

4.2.3 REPORTING

This Solution Element Virtual Interlocking OBB Phase 1 (VIL OBB 1.0.0) consists of two project ID's. The used metrics to control the engineering process are identified in the two different Engineering Dashboards. Several Components / Work packages listed in Table 4 and management activities listed in Table 3 are managed under a separate Project ID. The mapping can be found in the following Table 5: WBS mapping to Project IDs. As a result the reporting can be found under the following Engineering Dashboards:

Table 24: Mapping of Engineering Dashboard to Project ID

Project ID	Engineering Dashboard reference
100025795 / ATRELDE91	[EngDB SE VIL]
100020889 / ATRENDDB35	[EngDB EL BASE & TA]

4.3 CRITICAL TECHNOLOGIES

Not applicable in this solution.

4.4 RISK AND OPPORTUNITIES CONTROL

The Risk and Opportunities Control in this Solution will follow the Management System Process Manage Risks and Opportunities [S7] and is monitored in the [RiskReport].

4.5 CORRECTIVE ACTION PROCESS

Information about the Corrective Action Process can be found in the [QMP DIL].

4.6 INDEPENDENT EVALUATION AND VERIFICATION

Independent evaluation and verification will be done by

1. the verifier and domain experts upon instruction of the verifier (VER);
2. the validator (VAL);
3. the safety assurance manager (SAM).
4. The security assurance manager (SecAM) independently checks the project security manager (PSM) output.

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4.7 ESTIMATE/RE-ESTIMATE

The initial estimation is either an input for the phase “Orientation the Solution” or is done or verified during this phase and can be found in [DoP]. Re-estimation can be triggered by the SEM or GEN SEM typically after a WPM is raising major concerns reviewing the remaining work during the progress of working packages. The method used for estimation is mainly “Expert estimations” based on past experience, educated assumptions and risk considerations.

4.8 PLANNING

Planning is done in the tool PCM and can be found there with the identification listed in Table 1. During Bid phase initial cost plan is developed based on the provided features and Proposed Change Requests (PCRs). The planned values were consolidated in PCM in the forecast project. Latest by SOR2 see section [1.3.1] and based on Cost Breakdown Structure (CBS) are transferred into the tool PCM and can be found there with the identification listed in Table 1.

For planning of the product increments, epics, user stories and sprints described in [TeamCharterSEA] & [TeamCharterSEVIL] JIRA Advanced Roadmap is used following Agile Definition [S25].

For the interaction on PCM to the JIRA EPICs, each EPIC is bound to a specific account in PCM, see Tailoring of the monitoring of project progress within an agile development environment section [5.6]

4.9 PEER REVIEWS

The Peer Reviews in this Solution will follow the Minimum Requirements for the configuration management implementation [S9]. An overview of mandatory Documents and Reviewer has to be done according to Document List and Review Matrix for engineering projects [S23].

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5. DEVELOPMENT SCENARIO AND TAILORING OF THE SYSTEM ENGINEERING PROCESS

This Solution Element Virtual Interlocking OBB Phase 1 (VIL OBB 1.0.0) follow a hybrid approach.

Orient the Solution section 5.2.1, and Validate the solution section 5.2.5 follows the initialisation by Management System process Develop and Qualify the Solution (DDQS) as described in Project Life Cycle [S20].

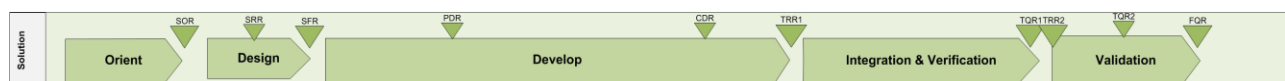
Work Packages

- Work Package ECMUX Phase 1
- Work Package CAE Data Tools Phase 1
- Work Package VIL Testing Phase 1
- Work Package ELEKTRA Base Phase 0 & Phase 1 - Component GENSYS_2.20.0
- Work Package ELEKTRA Base Phase 0 & Phase 1 - Component CAR-RED_1.0.0
- Work Package ELEKTRA ÖBB Phase 1

have tailored Design the Solution section 5.2.2 & Integrate and Verify the Solution section 5.2.4 follows the initialisation by Management System process Develop and Qualify the Solution (DDQS) as described in Project Life Cycle [S20] but have additional check by the agile development method applied in Engineering for small and medium sized projects as described in Agile Definition SMP [S25]. Develop the Solution section 5.2.3, follows only the initialisation by Management System process agile development method applied in Engineering for small and medium sized projects as described in Agile Definition SMP [S25].

TBD: Description of the tailoring in to Work Package CAE Data Tools Phase 1, Work Package ELEKTRA Test Automation Phase 0 & Phase 1, Work Package Test Automation Framework Phase 1, Work Package Test Automation Framework Converter

Figure 5: Develop and Qualify the Solution (DDQS)



Agile Definition SMP [S25] describes the setup, execution and monitoring of projects within Engineering, which have adopted an agile method for development. Benefits and Preconditions to the Agile Development expected benefits of agile development

- More flexibility with the handling of customer requirements
- More flexibility with the handling of changing priorities
- A significantly reduced time to test report for the System Test and associated rework
- Economic handling of late features or/and scope changes during the development process

A detailed guideline and tailoring to adapt the need of Solution Element Virtual Interlocking OBB Phase 1 (VIL OBB 1.0.0) is needed on team level.

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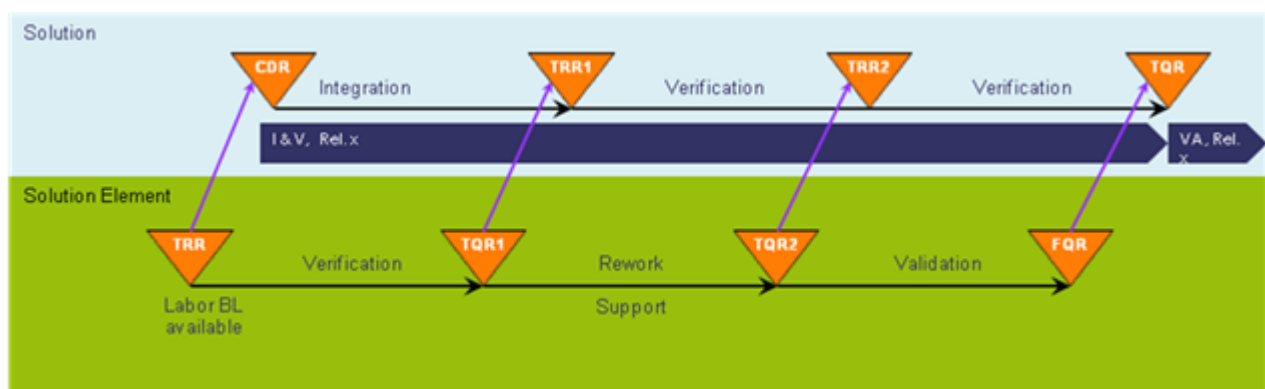
The following Work Packages using for this additional [TeamCharterSEVIL] deviations:

- ELEKTRA Base Phase 0 & Phase 1 - Component GENSYS_2.20.0
- ELEKTRA Base Phase 0 & Phase 1 - Component CAR-RED_1.0.0
- ELEKTRA ÖBB Phase 1
- VIL Testing Phase 1

Work Package ECMUX Phase 1 using additional deviation described in [TeamCharterSEA].

5.1.1 INTERACTION OF SOLUTION AND SOLUTION ELEMENTS

Figure 6: Interaction of solution and solution elements



5.2 TAILORING FOR THIS SOLUTION (DDQS PHASES)

5.2.1 ORIENT THE SOLUTION

No other deviation to the description in Project Life Cycle [S20].

5.2.2 DESIGN THE SOLUTION

No other deviation to the description in Project Life Cycle [S20].

The following Work Packages using Definition of Specification Done EPICs Checkpoints against template of Template of DoX Templates [S32] apply the agile development method applied in Engineering for small and medium sized projects as described in Agile Definition SMP [S25]:

- ELEKTRA Base Phase 0 & Phase 1 - Component GENSYS_2.20.0
- ELEKTRA Base Phase 0 & Phase 1 - Component CAR-RED_1.0.0
- ELEKTRA ÖBB Phase 1
- ECMUX Phase 1
- VIL Testing Phase 1

Todo: Deviations in Test Automation Framework Phase 1, Test Automation Framework Converter, ELEKTRA Test Automation Phase 0 & Phase 1

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5.2.3 DEVELOP THE SOLUTION

No Milestones PDR, CDR as deviation to the description in Project Life Cycle [S20].

The following Work Packages using Definition of EPICs Done checkpoints against template of Template of DoX Templates [S32] apply the agile development method applied in Engineering for small and medium sized projects as described in Agile Definition SMP [S25]:

- ELEKTRA Base Phase 0 & Phase 1 - Component GENSYS_2.20.0
- ELEKTRA Base Phase 0 & Phase 1 - Component CAR-RED_1.0.0
- ELEKTRA ÖBB Phase 1
- ECMUX Phase 1
- VIL Testing Phase 1

Todo: Deviations in Test Automation Framework Phase 1, Test Automation Framework Converter, ELEKTRA Test Automation Phase 0 & Phase 1

5.2.4 INTEGRATE AND VERIFY THE SOLUTION

The following Work Packages using Definition of Integration / Final Verification Epic Done checkpoints against template of Template of DoX Templates [S32] apply the agile development method applied in Engineering for small and medium sized projects as described in Agile Definition SMP [S25]:

- ELEKTRA Base Phase 0 & Phase 1 - Component GENSYS_2.20.0
- ELEKTRA Base Phase 0 & Phase 1 - Component CAR-RED_1.0.0
- ELEKTRA ÖBB Phase 1
- ECMUX Phase 1
- VIL Testing Phase 1

Todo: Deviations in Test Automation Framework Phase 1, Test Automation Framework Converter, ELEKTRA Test Automation Phase 0 & Phase 1

5.2.5 VALIDATE THE SOLUTION

The Assessment Certification according to CENELEC is needed for this solution element solution. No other deviation to the description in Project Life Cycle [S20].

Todo: Cybersecurity certification for this solution element.

5.3 TAILORING OF DOCUMENTATION MANAGEMENT

The tailoring of the Product Line Engineering (PLE) is used as described in Project Life Cycle [S20]:

- The naming convention for the name “Firstname Lastname” of the author, verified by, approver of a document can also be “F. Lastname”.
- Each document has to be trackable over a ticket in iTask, with a definition of done US like described in [TeamCharterSEA] & [TeamCharterSEVIL].

No other deviation to the description in Document Management Rules [S21].

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5.4 TAILORING OF MILESTONES

5.4.1 TAILORING OF DDQS MILESTONES

This describes deviations to the Project Life Cycle [S20] Milestones:

Milestone	Notes
Solution Orientation Review 1 (SOR1)	Intermediate Milestone to show the progress of SOR like to the description in Project Life Cycle [S20]
Solution Orientation Review 2 (SOR2)	No deviation to the description in Project Life Cycle [S20]
Solution Requirement Review (SRR1)	Intermediate Milestone to show the progress of SRR like to the description in Project Life Cycle [S20]
Solution Requirement Review (SRR2)	- No deviation to the description in Project Life Cycle [S20] - No deviation to the description in Agile Definition SMP [S25]. - After input data [SSS DIL], [SSS SEC DIL], [SecTRA], [MCRD DIL], [MCRD OBB], [SSDD DIL] released. More details on input data delivery date see chapter [3.4]. For Interaction of requirements solution and solution elements see [RMP] & process see chapter [5.1.1]. - After MS E03 Pflichtenheft Freigabe from Customer OBB defined in [PMP]. For latest available dates see [IMP]. - Work Package CAE Data Tools Phase 1 SRSSs, RTM, and VoR are not in scope of this milestone.
Solution Functional Review (SFR)	- No deviation to the description in Project Life Cycle [S20] - No deviation to the description in Agile Definition SMP [S25]. - After MS E03 Pflichtenheft Freigabe from Customer OBB defined in [PMP]. For latest available dates see [IMP]. - Precondition is [SSDD DIL] - Following work packages must perform a Definition of Specification Done to all EPICs of type specification against Template of DoX Templates [S32] a) Work Package ELEKTRA Base Phase 0 & Phase 1 - Component GENSYS_2.20.0 b) Work Package ELEKTRA Base Phase 0 & Phase 1 - Component CAR-RED_1.0.0 c) Work Package ELEKTRA ÖBB Phase 1 d) Work Package ECMUX Phase 1

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	e) Work Package VIL Testing Phase 1 Todo: Also apply to - Test Automation Framework Phase 1, Test Automation Framework Converter ELEKTRA Test Automation Phase 0 & Phase 1?
Preliminary Design Review (PDR)	Not used. - Deviation from Project Life Cycle [S20]. Instead Agile Definition SMP [S25] will be followed. See mapping of PDR in corresponding Agile Definition SMP [S25].
Critical Design Review (CDR)	- Not used - Deviation from Project Life Cycle [S20]. Instead Agile Definition SMP [S25] will be followed. See mapping of PDR in corresponding Agile Definition SMP [S25]. - No deviation to the description in Agile Definition SMP [S25]. - Following work packages must perform Definition of EPIC Done against Template of DoX Templates [S32] to all EPICs of type Design: a) Work Package ELEKTRA Base Phase 0 & Phase 1 - Component GENSYS_2.20.0 b) Work Package ELEKTRA Base Phase 0 & Phase 1 - Component CAR-RED_1.0.0 c) Work Package ELEKTRA ÖBB Phase 1 d) Work Package ECMUX Phase 1 e) Work Package VIL Testing Phase 1 Todo: Also apply to - Test Automation Framework Phase 1, Test Automation Framework Converter ELEKTRA Test Automation Phase 0 & Phase 1?
Test Readiness Review (TRR1)	- Deviation to the description in Project Life Cycle [S20], TRR is split into TRR1 for each Component/ Work Package see section [1.3.1]. - No deviation to the description in Agile Definition SMP [S25]. - Following work packages must perform Definition of EPIC Done to all EPICs of type Development & perform Definition of Test Epic Done to all EPICs of type Test against Template of DoX Templates [S32]: a) Work Package ELEKTRA Base Phase 0 & Phase 1 - Component GENSYS_2.20.0 b) Work Package ELEKTRA Base Phase 0 & Phase 1 - Component CAR-RED_1.0.0 c) Work Package ELEKTRA ÖBB Phase 1 d) Work Package ECMUX Phase 1

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	Work Package VIL Testing Phase must perform Definition of EPIC Done to all EPICs of type Development against Template of DoX Templates [S32]". Todo: Also apply to - Test Automation Framework Phase 1, Test Automation Framework Converter ELEKTRA Test Automation Phase 0 & Phase 1?
Technical Qualification Review 1 (TQR1)	- No deviation to the description in Project Life Cycle [S20] - No deviation to the description in Agile Definition SMP [S25]: - Work Package VIL Testing Phase must perform Definition of EPIC Test Done to all EPICs of type Test against Template of DoX Templates [S32]".
Test Readiness Review 2 (TRR2)	Together with TQR2. - Deviation from Project Life Cycle [S20].
Technical Qualification Review 2 (TQR2)	- No deviation to the description in Project Life Cycle [S20] - This Milestone is performed by Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0) - All Activities under Solution Element Virtual Interlocking Phase 1 Management & ELEKTRA BASE Phase 0 & Phase 1 Management in EPICs closed with Definition of Visibility Point Done against Template of DoX Templates [S32]
Final Qualification Review (FQR)	- This Milestone is performed by Generic Application Digital Interlocking OBB 1.0.0 (GA DIL OBB 1.0.0) - No deviation to the description in Project Life Cycle [S20]

5.4.2 TAILORING OF AGILE DEFINITION OF READY & DEFINITION OF DONE

This describes deviations to the Milestones: Template of DoX Templates [S32] apply the agile development method applied in Engineering for small and medium sized projects as described in Agile Definition SMP [S25]:

Agile Definition of Ready & Definition of Done	Notes
Visibility Point Ready	Not used

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Capability Ready	Not used
Specification Ready	Not used
Specification Done	ToDo Work package ECMUX have intermediate PreSWCRRs can be passed with NOK in the questionnaire and open action points, as long a SwCRR is reperformed on the Element
Epic Ready	Not used
UserStory Ready	Defined in [TeamCharterSEA] & [TeamCharterSEVIL]
UserStory Done	Defined in [TeamCharterSEA] & [TeamCharterSEVIL]
Epic Done	Work package ECMUX have intermediate PreSWCRRs can be passed with NOK in the questionnaire and open action points, as long a SwCRR is reperformed on the Element
Capability Test Ready	Not used
Capability Test Done	Work package ECMUX have intermediate PreSWCRRs can be passed with NOK in the questionnaire and open action points, as long a SwCRR is reperformed on the Element
Capability Done	ToDo
Visibility Point Test Ready	ToDo
Visibility Point Test Done	ToDo
Visibility Point Done	ToDo

5.5 TAILORING OF ROLE ASSIGNMENTS

5.5.1 ROLES OF AGILE

Solution Element Core Agile roles are described in section [3.1.2].

The following Work Packages apply the roles described in Agile Definition SMP [S25] and [TeamCharterSEVIL]:

- ELEKTRA Base Phase 0 & Phase 1 - Component GENSYS_2.20.0,
- ELEKTRA Base Phase 0 & Phase 1 - Component CAR-RED_1.0.0
- ELEKTRA ÖBB Phase 1
- VIL Testing Phase 1
- ELEKTRA Test Automation Phase 0 & Phase 1

Work package ECMUX Phase 1 apply the apply the roles described in Agile Definition SMP [S25] and [TeamCharterSEA] & [TeamCharterSEVIL].

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5.5.2 ROLE OF IVVQM

The role integration, verification, validation, qualification manager (short IVVQM) is not executed by a single person. The major IVVQM activities are distributed to other roles:

- Integration: lab integration is managed by the WPMs. Software integration testing is managed by the IVM
- Verification: this role is also distributed to other roles, see section [5.5.3]
- Validation: is managed by VAL
- Qualification: not needed

5.5.3 ROLE OF VERIFIER (VER) ACCORDING CENELEC EN 50716 AND EN 50129

The verification tasks will be handled according to CENELEC [S2] EN 50716:2024 – Railway application – Requirements for software development. The review of all documents is performed through reviews by the roles defined in the Document List and Review Matrix for engineering projects [S23]. The role VER is not executed by a single person. The responsibilities are defined in Table B.6 in [S2] and Table G.3 in [S6] (EN 50126-2:2017 – Railway Applications – The Specification and Demonstration of Reliability, Availability, Maintainability and Safety (RAMS) – Part 2: Systems Approach to Safety) :

- Process Verification: is managed by QAM
- Final document Verification: is made by VER or a person assigned by the SEM within the [DoP]
- Software/Code Verification: VER
- Deliverables Verification: is managed by IVM
- Enabling systems Verification: is managed by CM
- Lifecycle phases Verification: is managed by QAM
- Collection of Verification results in Verification-report: is made by IVM or a person assigned by the SEM within the [DoP]

5.5.4 ROLE OF DM

The role DM is not executed by a single person. The major DM activities are distributed to other roles:

- new ANV: can be obtained by DM
- Documentation Plan: updates are made by DM
- Document Index: is made by DM
- Formal check at release of a document: is made by QAM
- Other DM activities: are made by author of the document

5.6 TAILORING OF THE MONITORING OF PROJECT PROGRESS WITHIN AN AGILE DEVELOPMENT ENVIRONMENT

todo

Tailoring to the meeting. Refinement organized by the WPMs/POs.

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6. SYSTEM ENGINEERING SUPPORT

6.1 DATA AND CONFIGURATION MANAGEMENT

Data and Configuration Management is carried out according to Management System Process 9.7 CFG - Manage Configuration [S15], Document Management Rules [S21] and the specific Configuration Management Plan [CMP].

6.2 CONFIDENTIALITY

The rules as listed in Protection of Group Information [S18] apply.

6.3 EXPORT CONTROL

Export Control is carried out according to Management System Process 9.3 Control Export [S16].

6.4 CAPITALIZATION

Capitalization is handled in the context of the related Manage Product Policy process.

6.5 MEASUREMENTS AND ANALYSIS

See section 4.

6.6 TECHNICAL AND NON-TECHNICAL REQUIREMENTS MANAGEMENT AND CONTROL

For Requirement Management the tools DOORS and Polarion are used. The Requirements Manager ensures that requirements are

- uniquely identified
- unambiguous, testable and consistent
- traced from origin through all systems
- have all changes managed

The Requirements Manager manages changes throughout the life of the project. If requirements change, the impact will be reflected in the project plans, work products, and activities. Any inconsistencies between these customer- and/or system requirements and project plans, work products, and activities will be identified and resolved.

6.7 VALIDATION OF REQUIREMENTS

The validation is done in accordance with the Validation Qualification plans as listed in Documentation Plan [DoP].

6.8 DECISION ANALYSIS AND RESOLUTION

Decision analysis and resolutions are part of design and review processes. The results are documented in design documents, review reports and in this document and are listed in Documentation Plan [DoP]

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6.9 SUBORDINATE PLANS

Subordinate plans see

- Components / work packages see WBS managed as Work Packages section [1.1.4]
- Corresponding Input data plans mentioned in section [3.3].
- Corresponding plans created in Solution Element Deliverables section [1.3.2]

6.10 QUALITY ASSURANCE

The Quality Assurance in this Solution will follow Management System Guarantee quality and manage continuous improvement [S17]. Additional information can be found in the Quality Management plans as listed in Documentation Plan [DoP].

6.11 CERTIFICATION SUPPORT ACTIVITY

See Safety Management plans as listed in Documentation Plan [DoP].

6.12 HSE PLAN

The local requirements of the HSE Management and the measures resulting from it for the environment and protection of the employee's health are described in the HSE Instruction [S24]. HSE Plan is listed in Documentation Plan [DoP].

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APPENDIX A CRITERIA FOR THE SELECTION OF TECHNIQUES AND MEASURES

With each technique or measure in the tables there is a requirement for each software safety integrity level (SIL). The requirements for software safety integrity levels 1 and 2 are the same for each technique. Similarly, each technique has the same requirements at software safety integrity levels 3 and 4. These requirements can be

“**App**” this column selects with “**Yes**” for all relevant "documentation", “technique” or “measure” for the actual project or “**No**” if this row isn’t mandatory for the project.

‘**M**’ this symbol means that the use of a technique is mandatory,

‘**HR**’ this symbol means that the technique or measure is Highly Recommended for this safety integrity level. If this technique or measure is not used then the rationale for using alternative techniques shall be detailed in the Software Quality Assurance Plan or in another document referenced by the

Software Quality Assurance Plan

‘**R**’ this symbol means that the technique or measure is recommended for this safety integrity level. This is a lower level of recommendation than an ‘HR’ and such techniques can be combined to form part of a package,

‘**-**’ this symbol means that the technique or measure has no recommendation for or against being used,

‘**NR**’ this symbol means that the technique or measure is positively “Not Recommended” for this safety integrity level. If this technique or measure is used then the rationale behind using it shall be detailed in the Software Quality Assurance Plan or in another document referenced by the Software Quality Assurance Plan.

Table A. 1 Lifecycle Issues and Documentation

DOCUMENTATION	App	Basic Integrity	SIL 1	SIL 2	SIL 3	SIL 4	References of Doc Type
<i>Planning</i>							
1. Software Quality Assurance Plan		HR	HR	HR	HR	HR	[QMP DIL]
2. Software Planning and Verification Report		R	HR	HR	HR	HR	VER REP
3. Software Configuration Management Plan		HR	HR	HR	HR	HR	[CMP]
4. Software Verification Plan		HR	HR	HR	HR	HR	[IVP]

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5. Software Validation Plan		HR	HR	HR	HR	HR	[VQP]
Software requirements							
6. Software Requirements Specification		HR	HR	HR	HR	HR	SSS
7. Overall Software Test Specification		HR	HR	HR	HR	HR	TCS
8. Software Requirements Verification Report		R	HR	HR	HR	HR	VER Rep
Architecture and design							
9. Software Architecture Specification		R	HR	HR	HR	HR	SSDD
10. Software Design Specification		R	HR	HR	HR	HR	SSDD
11. Software Interface Specifications		HR	HR	HR	HR	HR	IFS
12. Software Integration Test Specification		R	HR	HR	HR	HR	TCS
13. Software/Hardware Integration Test Specification		R	HR	HR	HR	HR	TCS
14. Software Architecture and Design Verification Report		R	HR	HR	HR	HR	VER REP
Component Design							
15. Software Component Design Specification		-	HR	HR	HR	HR	SDD
16. Software Component Test Specification		-	HR	HR	HR	HR	TCS
17. Software Component Design Verification Report		-	HR	HR	HR	HR	VER REP
Component Implementation and Testing							
18. Software Source Code and supporting documentation		HR	HR	HR	HR	HR	Git
19. Software Component Test Report		-	HR	HR	HR	HR	Test Report

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20. Software Source Code Verification Report		-	HR	HR	HR	HR	VER REP
Integration							
21. Software Integration Test Report		R	HR	HR	HR	HR	Test Report
22. Software/Hardware Integration Test Report		R	HR	HR	HR	HR	Test Report
23. Software Integration Verification Report		R	HR	HR	HR	HR	VER REP
Overall Software Testing / Final Validation							
24. Overall Software Test Report		HR	HR	HR	HR	HR	Test Report
25. Software Validation Report		HR	HR	HR	HR	HR	VAL REP
26. Tools Validation Report		-	HR	HR	HR	HR	
27. Release Note		HR	HR	HR	HR	HR	VDD
Systems configured by application data							
28. Application Requirements Specification		HR	HR	HR	HR	HR	N/A
29. Application Preparation Plan		R	HR	HR	HR	HR	N/A
30. Application Test Specification		HR	HR	HR	HR	HR	N/A
31. Application Architecture and Design		R	HR	HR	HR	HR	N/A
32. Application Preparation Verification Report		-	HR	HR	HR	HR	N/A
33. Application Test Report		HR	HR	HR	HR	HR	N/A
34. Source Code of Application Data		HR	HR	HR	HR	HR	N/A
35. Application Data Verification Report		R	HR	HR	HR	HR	N/A

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36. Application Integration Test Specification		HR	HR	HR	HR	HR	N/A
37. Application Integration Test Report		HR	HR	HR	HR	HR	N/A
38. Application Release Note		HR	HR	HR	HR	HR	N/A
Software deployment							
39. Software Deployment Manual		R	HR	HR	HR	HR	N/A
40. Deployment Records		R	HR	HR	HR	HR	N/A
41. Deployment Verification Report		R	HR	HR	HR	HR	N/A
Software maintenance							
42. Software Maintenance Plan		R	HR	HR	HR	HR	N/A
43. Software Change Records		HR	HR	HR	HR	HR	N/A
44. Software Maintenance Records		R	HR	HR	HR	HR	N/A
45. Software Maintenance Verification Report		R	HR	HR	HR	HR	N/A
Software assessment							
46. Software Assessment Plan		-	HR	HR	HR	HR	N/A
47. Software Assessment Report		-	HR	HR	HR	HR	N/A
For Basic Integrity, the content of the software verification plan can be limited to 6.2.4.9 a), b), c), d), f). For Basic Integrity the software interface specification is only highly recommended for the boundary of the overall software (see also 7.3.4.18).							
NOTE 1 According to 5.3.2.12 and 5.3.2.13, documents can be combined differently. NOTE 2 Documents 29, 30 and 31 being HR or R depends on the importance defined in the process and where the verification takes place, e.g. data can only be needed to be imported but tested in the system domain while more functional properties need both test and verification. In this case HR has been marked but can be optional R. NOTE 3 Software Architecture Specification and Software Design Specification for Basic Integrity is important to establish especially for long-term maintenance, selection can consider this (e.g. adding this information within Software Requirements specification).							

Table A. 2 Software Requirements Specification

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TECHNIQUE/MEASURE	Ref	Basic Integrity	SIL 1	SIL 2	SIL 3	SIL 4
1. Modelling	Table A.17	R	R	R	HR	HR
2. Structured methodology	D.52	R	R	R	HR	HR
3. Decision Tables	D.13	R	R	R	HR	HR
Requirements: 1) The Software Requirements Specification shall include a description of the problem in natural language and any necessary formal or semiformal notation. 2) The table reflects additional requirements for defining the specification clearly and precisely. One or more of these techniques shall be selected to satisfy the Software Integrity Level being used.						

Table A. 10 Software Maintenance

TECHNIQUE/MEASURE	Ref	Basic Integrity	SIL 1	SIL 2	SIL 3	SIL 4
1. Impact Analysis	D.32	R	HR	HR	M	M
2. Data Recording and Analysis	D.12	HR	HR	HR	M	M

Techniques and measures according to EN 50129:2018 + AC:2019 Railway applications – Communication, signalling and processing systems – Safety related electronic systems for signalling:

Table E. 2 Safety Requirements Specification

TECHNIQUE/MEASURE	SIL 1	SIL 2	SIL 3	SIL 4	
1. Graphical description, e.g. block diagrams	R	R	HR	HR	
2. Structured Specification	HR	HR	HR	HR	(a)
3. Formal or Semiformal Specification	-	-	HR	HR	(b)
4. Computer Aided Specification Tools	-	R	R	R	(c)
Approved combinations of techniques for SIL 3 and SIL 4 are: - 1 and 2, or - 2 and 3 (in case of semiformal specification), or					

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- 3 (in case of formal specification).	
(a)	In order to reduce complexity, it is recommended that a structured requirements specification is developed through a hierarchical structure of refined requirements. This technique structures the functional specification into partial requirements such that the simplest possible, visible relations exist between the latter. This analysis is successively refined until small clear requirements can be distinguished. The result is a hierarchical structure of requirements which provide a framework for the specification of the complete black-box functional and safety requirements. This method emphasizes the interfaces of the requirements and is particularly effective for avoiding interface specification errors.
(b)	Formal or semi-formal specification provides a means of developing a description of a system based on a structured notation during the specification phase. Formal specifications are based on mathematical models of the system function and/or structure. Formal specification generally includes a verification process based on mathematical analyses to ensure the consistency of the resultant specification model. Semi-formal specification is recommended to express parts of a specification unambiguously and consistently, so that some mistakes, omissions and wrong behavior can be detected. For semi-formal specification, in some cases the description can be analyzed by machine or animated to display various aspects of the system behavior. For formal and semiformal methods see EN 61508-7:2010.
(c)	Model oriented procedures with hierarchical subdivision, description of all objects and their relationship, common database, and automatic consistency check.

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